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Summary / Resumen

This Technical Report summarizes the basic criteria used for the Global Finite Element Model idealization of aeronautical structures in Military A/C, used to support Static, Fatigue and Structural Dynamics analysis.

The guidelines presented are general criteria that can be updated depending on the particular characteristics of a structure, being focused on linear general models whose objectives are to reproduce real structure load path and obtain internal forces distribution across structural components.

The mission of these models will be obtaining the internal loads distribution across the structure and reproduce the load paths. The idealization should satisfy the requirements for Stress, Fatigue and Dynamic linear Finite Element Models.

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1. INTRODUCTION

1.1 Scope

The idealization by finite elements of a real structure consists in its conversion into an approximate mathematical model. This conversion process determinates the validity and usefulness of the results of the solution obtained, so that accuracy of the results are as good as the model is.

The practice of idealization is not documented in the technical literature, being the engineer's experience and company background the fundamental basis of the idealization work.

The basic tools can be learned in textbooks (See Ref. 2, Ref. 3, Ref. 4 and Ref. 5), but the capacity for idealization is acquired through years of experience. It is not therefore intended here to make a description of available elements or introduce the user to the elementary theory. The aim of this document is to establish basic modelling criteria focused on GFEM that condense the Airbus DS experience and homogenize the realization of structural finite element models.

In this procedure, the main basic criteria for the idealization of aeronautical structures to be applied in Static, Dynamic and Fatigue for the realization of general structural models (Global FEM, or GFEM) are established. These are basic criteria that can be updated depending on the particular characteristics of a model, being mainly aimed at linear and non-linear general models whose objectives are to obtain internal loads distribution across the structure and reproduce the load paths and stiffness, which will be used for validation of the strength of the structure (see Ref. 9). The idealization will first satisfy the requirements of Static and Fatigue for linear models, usually more detailed than the one required by Dynamics, but at the same time it will fulfil the requirements so that the model can be used directly by all, as well as with the basic rules for that with few or no changes can be used in non-linear solutions.

The current Guidelines apply to MSC.NASTRAN GFEM, but most of the criteria can be applied to OptiStruct or Lagrange GFEMs, with small modification or adaptations.

This Technical report has collected the contents from DPI-ADC-006 and will be used as a base for each project GFEM Model Guidelines that will be created for each A/C project.

1.2 Abbreviations And Symbols

Symbol	Units	Definition
A/C		Aircraft
BAI		Bending After Impact
BVID		Barely Visible Impact Damage
CAI		Compression After Impact
CoG		Center of Gravity
DFEM		Detailed Finite Element Model
DMM		Dynamic Master Model
DOF		Degree Of Freedom
e		Ultimate strain
Et	MPa	Traction Tensile Young Modulus
Ec	MPa	Compression Young Modulus
Ftu	MPa	Tensile Ultimate stress
Fty	MPa	Tensile Limit stress
Fcy	MPa	Compression Limit stress
Fsu	MPa	Shear Ultimate stress
G, Gii	MPa	Shear Modulus
GFEM		Global Finite Element Model
GVT		Ground Vibration Test
MPC		Multi Point Constraint
RBE		Rigid Bar Element
SMM		Structural Master Model
SPC		Single Point Constraint
TAI		Tension After Impact
VID		Visible Impact Damage
μ, μ_{ii}		Poisson ratio

1.3 References

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	DPI-ADC-007	F	"Global Finite Element Model Verification And validation Procedure".
Ref. 2	J. ROBINSON, ROBINSONS AND ASSOCIATES, 1981		"UNDERSTANDING FINITE ELEMENT STRESS ANALYSIS"
Ref. 3	O. C. Zienkiewicz and R. L. Taylor, McGraw-Hill Book Company, Fourth Edition		"The Finite Element Method".
Ref. 4	F. Hartmann – C. Katz, Springer, 2004		"Structural Analysis with Finite Element".
Ref. 5	F.L. Matthews, G. A. O. Davies, D. Hitchings and C. Soutis		CRC Press, 2000 "Finite Element Modelling of Composite Materials and Structures".
Ref. 6	MSC.Software		"MSC.Nastran Quick Reference Guide".
Ref. 7	MSC.Software		MSC.Nastran "Getting Started with MSC.Nastran User's Guide"
Ref. 8	Airbus DS Design Manual. Volume 8. Chapter 152		
Ref. 9	DP-ADC-002	U	"Validación de resistencia estática de la estructura".
Ref. 10	35-D-551 80022		"Torque box spars. F.E.M. models".
Ref. 11	P. Kuhn, McGraw-Hill Book Company, 1956		"Stresses in Aircraft and Shell Structures".
Ref. 12	MSC.Software		"MSC.Nastran Superelement User's Guide".
Ref. 13	FM0200016	1.0	"A350 XWB HTP & VTP GFEMS: GUID"
Ref. 14	NT-T-ADP-98011	A	"Proc. de obtención de características equiv. de rigidez de paneles con aberturas".
Ref. 15	NT-5-ADS-98003	A	"Modelos de Elementos Finitos de los 'Manholes' del Ala del C295", 1998.
Ref. 16	M.C. Y. Niu	2nd	"Airframe Stress Analysis and Sizing"
Ref. 17	ESDU 84092		"Stiffness Data for Finite Element Calculations Equivalent Stiffness Factors for Shear Webs", 1984
Ref. 18	NT-T-ADP-04002	A	"Generación de Librerías PBARL para MSC.NASTRAN", 2004
Ref. 19	C/MC/84-184		"Modelización de paneles sándwich en Materiales compuestos", 1984
Ref. 20	Sang H. Lee, MSC.Software		MSC.Nastran "Handbook for Non-Linear Analysis" (Draft)
Ref. 21	NT-0-ADP-08001	A	"Guidelines for Static and Dynamic FEM Validation by Ground Vibration Test and Structural Test Results Analysis", 2008

2. GFEM MODELLING PRINCIPLES

2.1 GFEM Modelling Process

The process of creating a global finite element model requires the execution of a series of steps common to all models:

1. Understand the way the real structure works. It is not possible to make a good finite element model without identify exhaustively the way of working of the structure that is being idealized. So, that means to identify:
 - How the structure is supported
 - How it is loaded
 - Which are the main load paths
 - What is the main structural function of each element
2. Build the finite element model itself.
 - Define coordinate systems
 - Define the nodes of the mesh to be created
 - Define the types and the properties of the elements to be used. This point is the determinant of the type of model performed and will have greater importance in the rest of the procedure.
 - Define the type of support constraints and the interfaces of the model
 - Define the load application method and elements
3. Verify the model.
 - According to Ref. 1
4. Run the analysis and obtain FE results. The official GFEM should be stored at a specific location and cannot be modified without required permissions.
5. Register, document and archive the model.
 - It is done as described in Ref. 1. In this reference is described the identification to be given to the model.
6. If the FE Model is required to be updated or modified, modification should be tracked and controlled, GFEM shall be re-verified and steps 3-5 reproduced with a new version of the GFEM.

In fact, many of the previous aspects of the idealization process must be considered simultaneously since they only have meaning in relation to the rest of the aspects and depending on the objective of the model. Therefore, the procedure will focus on the different types of elements to be used in the general models of the most common structures, highlighting the main hypotheses implied by the use of each one.

By functional organization of the model, it is very useful to subdivide the whole model into groups, so that each zone of the model or type of data is grouped into a file. The general model will be formed by the union of all the previous files and will only be formed in the execution. This joining process can be done through a command procedure or using MSC.Nastran INCLUDE card (Ref. 6).

2.2 GFEM Units

GFEM used for Structural Analysis (also known as SMM or Structural Master Model) and Dynamic Analysis (also known as DMM) will be performed in the following units, unless other agreement exist for the program:

	Structural Master Model Units	Dynamic Master Model Units
Distance	Milimeters (mm)	meters (m)
Force	Newton (N)	Newton (N)
Mass	Tons (Mg) or Kilograms (Kg)	Kilograms (Kg)
Time	second (s)	second (s)

Table 1. GFEM Model units

If Mg is the masses unit for SMM, the units system for the GFEM is coherent.

If Kg is the masses unit, a factor should be used for obtaining the modes frequency values in Hertz (Hz) → PARAM, WTMAS, 1.E-03 (Ref. 6).

2.3 Numbering of Elements and Nodes

The numbering system used will be a cyclic numeration of the nodes with a constant increase that allows an easy modelling and checking as well as an easy identification of the output data and interpretation of results



Figure 1. HTP Box Elements Identification

As an example, in the Figure 1, GRIDS and Elements start with “57” that corresponds to HTP-BOX

- GRIDS Identification: 57XXYZZ,
 - XX= 16 or 20, corresponds to Rib position (Rib04 Rib05)
 - Y = 5 for upper skin
 - ZZ corresponds to Stringer ID. 00 corresponds to rear Spar, 02 corresponds to Str01...
 - Identification criteria allows to introduce intermediate nodes
- Elements Identification: 57EXXYZZ,
 - E= 1 for 1D stringer element, 2 for shell skin element

In most Military A/C programs, numbering criteria is established by components, so that integration and post-processing are simplified. If none is established, some criteria are proposed in this procedure, used in Airbus DS.

The numbering system proposed for the nodes consists of 7 digits where each one indicates a particular location according to the following scheme:

Component	GRID Identification Range	Element/Property/Material
Fuselage	1 000 000 → 4 999 999	10 000 000 → 49 999 999
VTP / Rudder	5 000 000 → 5 399 999	50 000 000 → 53 999 999
HTP / Elevator	5 400 000 → 5 999 999	54 000 000 → 59 999 999
Wing (Left)	6 000 000 → 6 999 999	60 000 000 → 69 999 999
Wing (Right)	7 000 000 → 7 999 999	70 000 000 → 79 999 999
Pylon & Engines, Landing Gear	8 000 000 → 8 999 999	80 000 000 → 89 999 999
Aeroelastic	9 000 000 → 9 999 999	90 000 000 → 99 999 999

Table 2. GRID and Element Identification Range

The numbering criteria shall be fixed and determined at the beginning of the program, according to the needs of each component regarding number of nodes/elements.

2.4 Material Cards

Usual Material cards used for GFEM are MAT1, MAT2 and MAT8.

FE modeler has the responsibility to choose the right Material properties to be used for each aircraft part or component, according to rules defined in following paragraphs. The Material data and used and the references shall be documented within GFEM description document.

2.4.1 Metallic Material Properties (MAT1)

- Moduli E and G (or Poisson ratio): Mean value between tension and compression,
- Base value: Mean values.
- Temperature: Use the critical (minimum) moduli values available for the corresponding working temperature

2.4.2 Composite Ply Properties (MAT8)

- Moduli E_i , G_{ii} and μ_{12} : Mean moduli values between tension and compression,
- Base: Mean values.

- Temperature and Ageing: Use the critical (minimum) moduli values available at the working temperature for the corresponding structure. Wet ageing for strength analysis. Room Temperature values for Full Scale Test GFEM Correlation.
- 0 deg Fiber orientation for shell elements (e.g. PCOMP/MAT8) has to be defined by a coordinate system (MCID and not by angle definition on PCOMP card).

2.4.3 Anisotropic Material Properties (MAT2)

- The Material card is based on defining Material matrix terms G_{ij}
- For composite laminates G_{ij} are obtained from Classic Laminate Theory using the stacking sequence.
- A MAT2 card is also used for metallic materials to represent Shear-Only panels (with G_{33} as relevant figure) or to adjust Torsional stiffness (4.1.7).

2.5 Structural Symmetry

The structural symmetry will only be used for linear behavior, never for non-linear models or for buckling solutions (linear or non-linear). This method can be used to reduce the dynamic analysis computation costs.

A linear behavior structure is symmetric regarding to a plane when it has the same geometry, supports, elements and properties of materials on both sides of the plane of symmetry, it is not necessary that the loads applied on each side are symmetrical regarding to the same plane.

The use of symmetry has the following advantages:

- Reduces the size of the structure that is being idealized.
- Reduces the work required to create the model and the amount of input data, although it complicates a bit the boundary conditions and the introduction of loads.
- Reduces the execution time of the model (currently this advantage is not very appreciable).

When a structure has symmetry plane and is loaded symmetrically (or anti-symmetrically) regarding to the plane of symmetry, the internal loads and deformations of a point in the structure and in its mirror image are the same (or the same with opposite sign). It is therefore enough to analyze the total structure to take into account the portion of structure at one side of the plane of symmetry.

The use of symmetry involves setting symmetric and antisymmetric support conditions, decomposing load cases into symmetric and antisymmetric components and finally superimposing stresses and strains to obtain the final result and therefore this possible structural symmetry will only be used in models that have approximately more than one hundred thousand degrees of freedom, or in those cases in which it is sufficiently justified its use, being completely discouraged in buckling problems or in non-linear calculations.

As it was explained in the previous paragraph, the structural symmetry can be used in a linear analysis even when the applied charges are non-symmetric

The process to be followed will be:

The applied loads must be decomposed into their symmetric and antisymmetric components, using the following expressions:

Symmetric load component in a node: $(F_{right} + F_{left})/2$

Antisymmetric load component in a node: $(F_{\text{right}} - F_{\text{left}})/2$

Where F_{right} and F_{left} are the forces of a load case in a node on the right side and in its symmetry on the left side respectively

Run the FE model with symmetrical contour conditions in the plane of symmetry and with the symmetric component of the applied loads. For all the nodes located in the plane of symmetry it will be prevented the displacement as well as the rotation around axes in the plane of symmetry.

Run the FE model with antisymmetric boundary conditions in the plane of symmetry and with the antisymmetric component of the loads applied. For all the nodes located in the plane of symmetry, it will be prevented the displacement within the plane as well as the rotation around an axis perpendicular to the plane of symmetry.

Superimpose the results of the two previous executions to obtain the final non-symmetric solution.

It should be noted that, in addition to the boundary conditions imposed in the plane of symmetry of the model, it will be necessary to constraint the three degrees of freedom that the structure presents as a rigid body (two translations in the plane of symmetry and one rotation around the perpendicular axis in the symmetrical case and a translation perpendicular to the plane and two rotations around axes in the antisymmetric case) in a statically determined way. That is, one and only one node of the plane of symmetry will have all its degrees of freedom constrained.

The following pictures show the components of translation and rotation in a symmetric structure with symmetric loading (Figure 1) and antisymmetric loading (Figure 2).

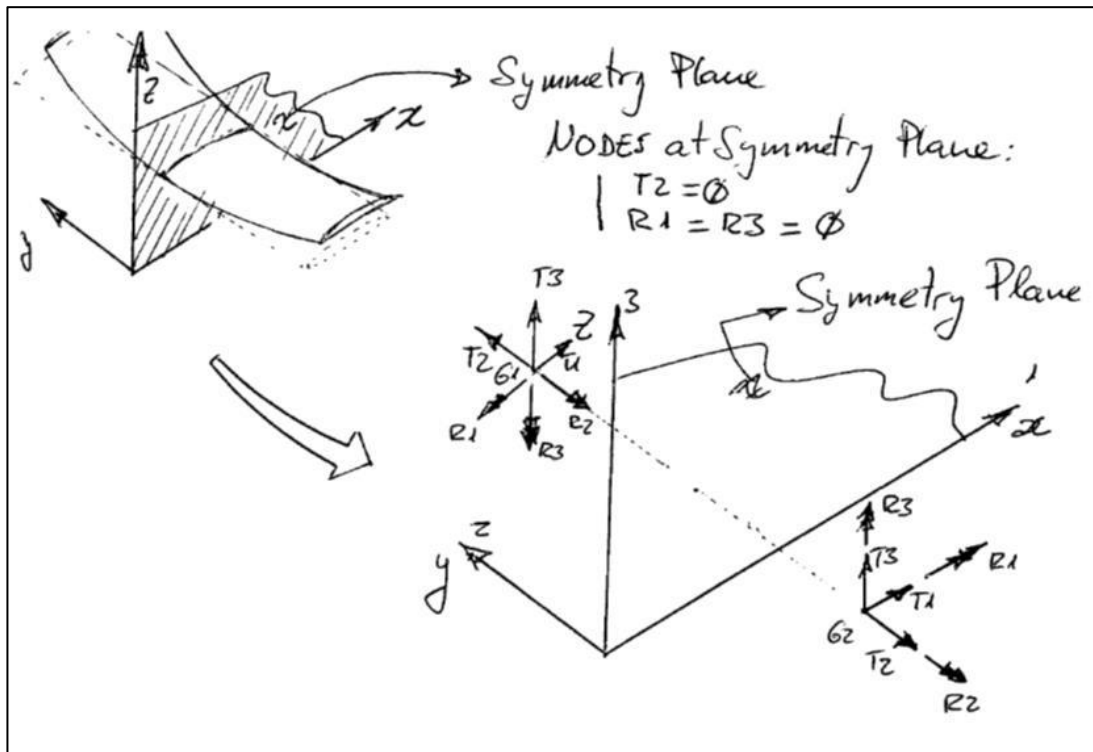


Figure 2. Symmetrical conditions

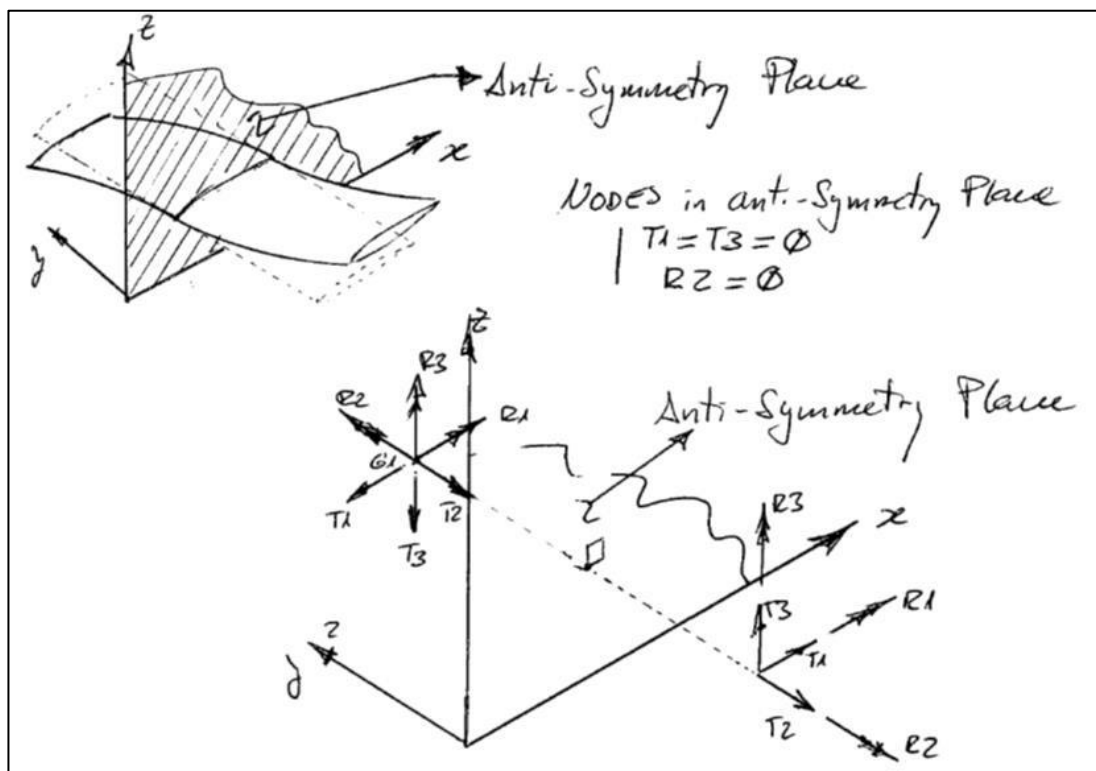


Figure 3. Anti-Symmetrical conditions

As can be observed, the symmetric load produces a symmetric distribution of displacements and antisymmetric of rotations, while an antisymmetric load induces an antisymmetric distribution of displacements and symmetric of rotations.

On the other hand, structural symmetry can also be used through the super-elements technique by means of what is called the image method (see Ref. 9). This method is mentioned in chapter 11.4 concerning the use of the super-elements technique.

In any case, when the structural symmetry is used in a component of the structure of the airplane, the right part of it will be idealized with the following axes criterion:

- X axis backwards of the flight direction.
- Y axis to the right (looking in the flight direction).
- Z axis upwards, giving a right-hand system.

The reason for this agreement lies in the fact that in planes the left part is drawn using the z axis downwards.

2.6 MSC.Nastran Rules for GFEM

The following rules must be followed in the creation of a GFEM model, using MSC.Nastran:

- The use of PARAM AUTOSPC in the BULK DATA (from MSC.NASTRAN 2005 version onwards, AUTOSPC can be a Case Control command, Ref. 6), the parameter PARAM, K6ROT (which establishes a value of a stiffness in the normal degree of freedom to the 2D elements) and SNORM, will be determined by the values specified in each program and for each specific version of NASTRAN and will be in accordance with the requirements of Ref. 8. As a general rule, for linear models used to obtain internal loads, it will be used: PARAM, AUTOSPC, NO; PARAM, K6ROT, 1.0 and PARAM, SNORM, 20.0
- Always request the output SPCFORCES = ALL, and verify in the output files (F06, PCH, HDF5, OP2 or XDB) that the reactions are located where expected (Ref. 1).
- The reference coordinates systems used in MSC.NASTRAN are right handed and their number must be as low as possible. In symmetric structures, the origin of the system and two axes will be in the plane of symmetry. Rectangular coordinate systems shall be provided whenever possible, and there are no reasons to justify the contrary, using CORD2R cards to detach them from possible posterior position changes of the reference nodes.
- Use the 'BASIC' displacement axes system whenever possible and explicitly remark those nodes whose axes system cannot be changed because this change would cause changes in the stiffness of the structure (for example in the interfaces or in the definition of 'offsets').

It should be remembered that in the MSC.Nastran bibliography, the BASIC coordinate system is the implicit reference system (the system enunciated by default for the definition of the nodes), while the GLOBAL system is the set of all the displacement systems existing in the model. The outputs concerning the nodes are always given in the displacement coordinate and the field of the GRID card where is defined is the field 7. The default value in case of leaving blank the field 7 is the basic system. Therefore, field 7 of the node definition card will be left blank, whenever possible.

- Structured solutions are desirable: SOL 101 for static linear analysis and SOL 103 for modal analysis.
- Use RBE3 cards for mass or load distribution between several nodes. However, the different use of these RBEx cards is subject to further analysis in chapter 10.2.
- It is noted that by using PARAM cards in the BULK DATA, the default output of a GPFORCE can be altered, consisting of values of the actions expressed in the global coordinate system, which can be very useful in many occasions. This is done by giving a negative integer value to the PARAM NOGPF card. With a positive integer value for the PARAM NOELOF card, the forces will output in the direction of the edges of the contiguous shell elements, and with a positive integer value for the PARAM NOELOP card the sum of the forces will appear in the direction of the edges of the adjacent elements.

In Ref. 6 can be found more information on this topic.

- The aspect ratio of the shell elements CQUAD4 must not exceed the value of 10 (elements with higher relations of 10 can be acceptable but will require additional checks and acceptance will be under responsibility of FEM specialist, see Ref. 1).

3. GFEM NODES AND CO-ORDINATE SYSTEMS

3.1 Nodes Location

The nodes will be placed on the outer surface of the skin, as this is the surface supplied by the CAD system and will not be linked to the type of elements to be used.

Unless other constraint, the nodes that define the mesh of the skin will be placed in the position of each stringer and in the intersection of the plane of the ribs or frames, so that the elements defining model of the panel will be between stringers and ribs (natural mesh).

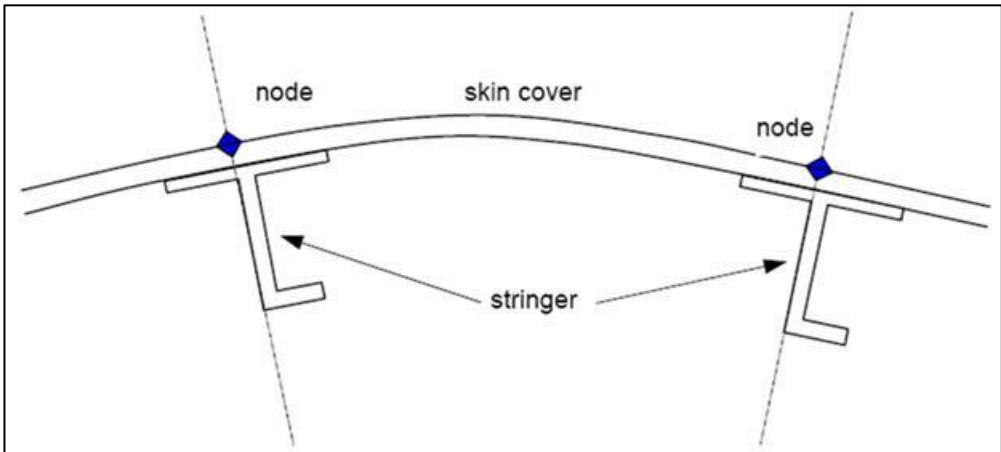


Figure 4. Skin nodes position

For Fuselage, Wings of Horizontal Tail Planes there will be nodes located at the A/C symmetry Axis, even in the case that no structural elements as wing ribs or fuselage stringers are defined in the symmetry plane.

3.2 GRID Card Definition

1	2	3	4	5	6	7	8	9	10
GRID	ID	CP	X1	X2	X3	CD	PS	SEID	

Table 3. NASTRAN Grid Card definition

- Use the 'BASIC' displacement axes system (CD, Field 7) whenever possible
- Explicitly remark those nodes with CD axes systems (both in GFEM documentation and with comments in NASTRAN Bulk data file). These CIDs cannot be changed because this change would cause changes in the stiffness of the structure (for example in the interfaces or in the definition of 'offsets').
- Nodal outputs are always given in the displacement coordinate and the field of the GRID card where is defined is the field 7. The default value in case of leaving blank the field 7 is the basic system. Therefore, field 7 of the node definition card will be left blank, whenever possible.
- No degrees of freedom of nodes should be restricted in field 8 of the GRD definition card, not even degrees of freedom in rotation, since it would make it impossible to use the model in dynamic analyses. SPC/SPC1 cards will be used for this purpose.

3.3 Coordinate Systems

GFEM Reference Coordinate System (CID=0) will be the A/C (or component) reference co-ordinate system used in DMU for the project, also known as ABCS (Aircraft Basic Coordinate System).

The majority of the GRID nodes will be defined in this ABCS (CID=0), let's say, CP field in GRID card (Field 3) will be "0" or blank.

Some other Co-ordinate Systems can be used :

- Third party models: Engines, Landing Gear
- Components suitable to be shifted

Specific local CID will be defined for specific purposes:

- Movable surfaces Hinge Lines
- Component Elastic Axes (see Chapter 10)
- Specific interfaces to get right interface loads
- Cylindrical CID for Fuselage offset or reference definition.

Material Coordinate Systems: Material coordinate systems will be defined to identify Fiber 0 deg orientation for each structural component. Different Material Coordinate systems will be defined for Right and left composite material references.

4. LIFTING SURFACES MODELLING

Common lifting surfaces structure is built by a stiffened skin (skin with stiffeners) or sandwich skin, and some spars and ribs.

4.1 Skin Modelling

The skins are structural elements that work mainly in compression-tension and shear in a flat stress state.

The slight bending through the thickness is due to the difference of coordinate regarding to the neutral line that exists between the interior and exterior points of the skin and, for this reason, negligible in this approach. Some exception to this hypothesis is in case of sandwich panels or thick skin panels (as the used in Fighter A/C), where this effect should be taken into account.

Different skin modelling are proposed, based on type of material used in manufacturing and the relative inertia of the stringer compared with the wing box, h/h' (see Figure 5), where:

- h : Distance between the outer surfaces of the upper and lower skins (distance of the nodes in the model).
- h' : Distance between the centers of gravity of the upper and lower coverings (panel plus stringers)

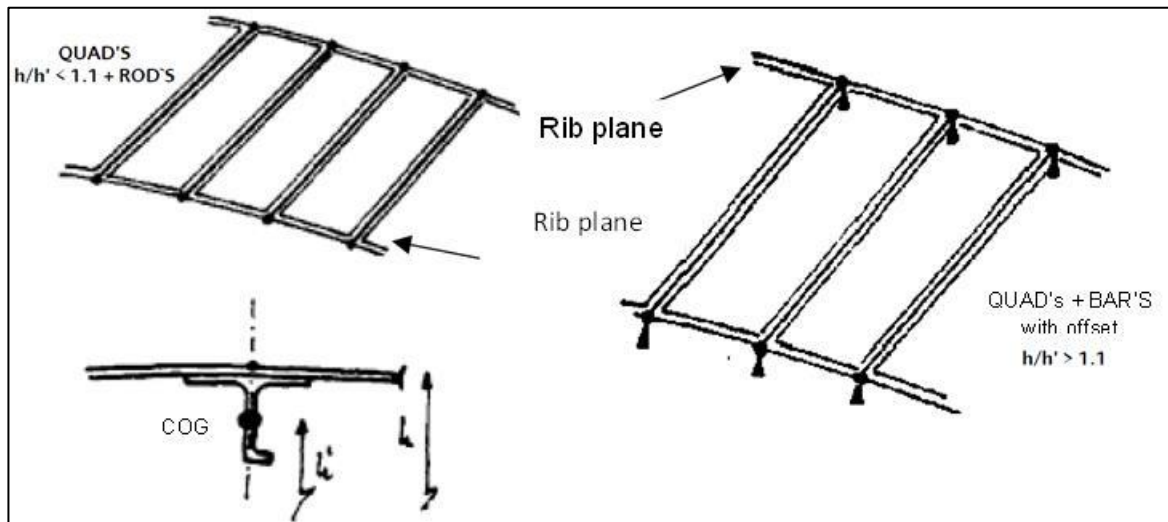


Figure 5. Basic Skin Modelling

4.1.1 Metallic Skin Modelling ($h/h' < 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties: PSHELL card with the thickness of the base skin and with membrane properties only (MID1, without bending capability- MID2=0).
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD elements
 - Properties: PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists.

- Material: MAT1 card with E modulus. Data extracted from Material database.
- Element Orientation:
 - Shell: X_{element} from root to tip, Z_{element} looking upwards, Y_{element} from Rear Spar to Front Spar (RHS) and in opposite direction in LHS. See attached Figure.
 - Stringer: X_{element} from root to tip.

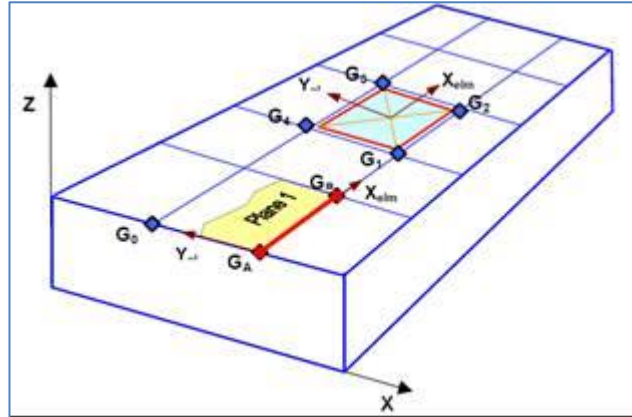


Figure 6. Wing / HTP Element Axes definition

4.1.2 Metallic Skin Modelling ($h/h' > 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties:
 - Thin Skin: PSHELL card with the thickness of the base skin and with membrane properties only (MID1, without bending capability- MID2=0).
 - Thick Skin: PSHELL card with the thickness of the base skin and with shell properties (membrane and bending). Offset will be required when element thickness 't' is $h/(h-2t) > 1.1$.
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD or CBAR elements
 - Properties:
 - PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists factorized with coefficient h/h' .
 - PBAR plus offset values, with the area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists, and corresponding Inertias.
- Material: MAT1 card with E modulus. Data extracted from Material database.
- Element Orientation:
 - Shell: X_{element} from root to tip, Z_{element} looking upwards, Y_{element} from Rear Spar to Front Spar (RHS) and in opposite direction in LHS. See Figure 6.
 - Stringer: X_{element} from root to tip.

4.1.3 Solid Laminate Composite Skin Modelling ($h/h' < 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties/Material: PSHELL or PCOMP cards

- PSHELL/MAT2: with the thickness of the base skin and with properties according to classic laminate theory.
- PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8.
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD elements
 - Properties/Material: PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists. MAT1 corresponding to equivalent isotropic modules in the direction of the stringers, according to Classical Laminate Theory..
- Element Orientation:
 - Shell: X_{element} from root to tip, Z_{element} looking upwards, Y_{element} from Rear Spar to Front Spar (RHS) and in opposite direction in LHS. See Figure 6.
 - Stringer: X_{element} from root to tip.

4.1.4 Solid Laminate Composite Skin Modelling ($h/h' > 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2: with the thickness of the base skin and with properties according to classic laminate theory.
 - PCOMP/MAT8: PCOMP card with real stacking sequence.
 - Offset will be required when element thickness 't' is $h/(h-2t) > 1.1$.
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD or CBAR elements
 - Properties/Material:
 - PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists factorized with coefficient h/h' .
 - PBAR plus offset values, with the area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists, and corresponding Inertias.
 - MAT1 corresponding to equivalent isotropic modules in the direction of the stringers, according to Classical Laminate Theory.
- Element Orientation:
 - Shell: X_{element} from root to tip, Z_{element} looking upwards, Y_{element} from Rear Spar to Front Spar (RHS) and in opposite direction in LHS. See Figure 6.
 - Stringer: X_{element} from root to tip.

4.1.5 Sandwich Composite Skin Modelling

Usual sandwich skin doesn't have stringers.

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2: with the thickness of the base skin and with properties according to classic laminate theory, including core.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8.
 - Offset required in CQUAD4 or PCOMP card.

- Element Orientation:
 - Shell: X_{element} from root to tip, Z_{element} looking upwards, Y_{element} from Rear Spar to Front Spar (RHS) and in opposite direction in LHS. See Figure 6.

4.1.6 Exceptions

- Inter-Rib Nodes: When the model has nodes between the ribs, it is necessary to use elements with stiffness out of the plane. This will be the case of a skin that has an idealized hole or a stringer trimmed between ribs or it is desired to take into account the effect of the internal pressure of fuel. In these cases it will be used CBAR elements (CBEAM for non-linear) without offset and with the appropriate bending stiffness (the corresponding to the stringer plus the area of extra thickness) instead of CROD elements. For thick skins and for obtaining more accuracy, the stiffness of the panel can be added to the stiffness of the CBAR element relative to its midline (adding the bending stiffness in the CQUAD4 element).
- Stringers with relevant inertia: When the stiffeners have an appreciable height regarding to the height of the torsion box (for example, they differ by an order of magnitude only) it is necessary to take into account explicitly the effect of the position of the center of gravity of the resistant area. If not, while the value of the internal force or stresses can be corrected later to take this effect into account (as described below), the error committed in the stiffness would be relative high and not easily correctable. In this case CBAR elements with offset will be used, being this the distance from the center of gravity of the stringer plus 'pad' to the reference line of the nodes. For non-linear studies, CBEAM elements without offsets will be used instead of CBARs with offsets, with the bending properties corrected to the outer point of the skin (reference line of the nodes).
- When the two previous situations occur simultaneously it will be used CBAR elements with offset or CBEAM without offset and corrected inertias whether it is a linear or non-linear study.

For the idealization of panels with holes (for example, "manholes"), it is recommended to follow the same criteria defined in for the holes in the spar, ribs or windows.

The use of CBAR/CBEAM elements means the introduction of a larger number of input data and thus a longer MSC.Nastran calculation time, so they will only be used in those cases in which the accuracy of the results requires it. The fact of placing the nodes of the model on the outer surface of the skin and because of the proposed idealization makes the bending stiffness of the model slightly higher than the stiffness of the real structure. Therefore, the results obtained from the model must be corrected before to carry out the strength check with them. Due to this fact, the internal axial forces and axial stresses of the two skins obtained from the model must be multiplied by the factor:

$$h/h' \text{ (see Figure 5)}$$

Being:

- h : Distance between the outer surfaces of the upper and lower skins (distance of the nodes in the model).
- h' : Distance between the centres of gravity of the upper and lower coverings (panel plus stringers)

The deflection of the nodes of each section obtained from the model must be affected from the previous square factor (if this factor is assumed constant for all sections) to obtain the deflection corrected for that effect.

In typical torsion boxes this factor does not exceed the value 1.05 (the stresses will be 5% higher than those obtained in the model). In cases where this factor reaches the value 1.1, errors in the distribution of load between elements of the section itself can be significant. In this case CBAR elements with offset will be used, being this the distance from the centre of gravity of the stringer plus 'pad' to the reference line of the nodes, and without torsional stiffness for linear models and CBEAM elements without offsets instead of the CBAR with offsets, with the bending properties corrected to the outer point of the skin (reference line of the nodes) and the rest as those of the CBAR of the linear execution.

The idealization proposed with CQUAD4 and CROD elements is the most suitable for stiffened panels (Ref. 10), in which there is no local buckling of the panel between stiffeners at a certain value of the applied load and therefore the stiffness of the skin remains constant regardless of the load level.

For the specific cases that the skin is a PANEL-COLUMN type with buckling of the panel between stringers at low levels of applied load, a non-linear effect of loss of stiffness of the skin occurs. The load that supports each skin is affected by this fact because the centre of gravity of the resistant area changes slightly as it decreases, approaching the middle line of the torsion box. This effect is usually very small and is not taken into account but it can be estimated conservatively considering that the strength of both skins has increased in the proportion

$$h'/h''$$

Being:

- h' : Distance between the centres of gravity of the upper and lower coverings (panel plus stringers)
- h'' : Distance between the centres of gravity of the upper and lower skins, considering for the compression skin only the effective area in compression due to the applied load.

For an extreme situation in which the idealized skin is formed by a very thin sheet stiffened by strong stringers, it would be convenient to idealize this sheet with CQUAD4 elements with only shear stiffness (introducing only G33 and occasionally, for dynamic reasons, it may be necessary to give small values to the components G11 and G22 with a MAT2) and add, to the area of the adjacent stringers, the effective area in compression of the skin panel (approximately 15 times the thickness on each side of the stringer). These stringers will be idealized with CROD elements. It should be noted that the load obtained on each skin is independent of the type of elements used to idealize the sheet and only depends on the position assigned to the centre of gravity of the assembly. The same does not occur with the deflections of the nodes for each section.

As a guide, Table 4 and Table 5 summarize the type of idealization to be followed in each case and the factor to be applied to the Nastran output for the correct interpretation of results.

	Stiffened panels				Panel-Column
	h/h'<1.1		h/h'>1.1		
	Without mid nodes	With mid nodes	Without mid nodes	With mid nodes	
Base Panel	CQUAD4		CQUAD4		CQUAD4(G ₃₃)
Stringers	CROD	CBAR w/o offsets	CBAR with offsets		CROD

Table 4. Skin modelling

Magnitude	Coefficient
Axial forces	h/h'
Axial stresses	h/h'
Vertical displacements	(h/h') ²

Table 5. Results interpretation: Coefficients for output values for skin

4.1.7 Torsional Stiffness correction

According to Ref. 13 and the results obtained in Ground Vibration tests, it is known that the current GFEM idealization of the lifting surfaces explained in previous sub-sections (with stringer represented by a CROD/PROD with no J contribution) predicts a torsional stiffness modeled structure that is conservatively lower than that of the real one, especially in the case of integrated skin structures (metallic machined or composite co-bonded or co-cured).

In the real structure, the stringer flanges and feet are carrying shear as well and thereby contributes to the lifting surface box torsional stiffness, which cannot be represented with the current idealization. Furthermore, in the GFEM idealization, each skin panel has its minimum thickness. In real design there is the effect of what is called “cliffedge”. This occurs when both panels at each side of a stringer have different thickness, with the stringer always located in the top of the thicker part.

This section outlines the methodology used to obtain a more accurate stiffness including the torsional stiffness of the lifting surface FEM to account for the stiffening effect for the stringer attach flange.

The theory permits the torsional stiffness (GJ) of a section of the structure to be calculated based on the end rotation. This end rotation is given by

$$\theta = \frac{TL}{4A^2G} \oint \frac{ds}{t}$$

where 'T' is the applied Torque, 'L' is the length of the box structure, 'A' is the enclosed area of the centre lines, 'G' is the Bulk Modulus or Shear Stiffness, 'ds' is the distance across the section and 't' is the thickness.

Taking into account the influence of the stringer flange and the cliff edges on the torsional stiffness of the box, the skin properties in the GFEM will be modified. The correction made for each panel due to the stringer flanges and cliff edges is depicted in Figure 7.

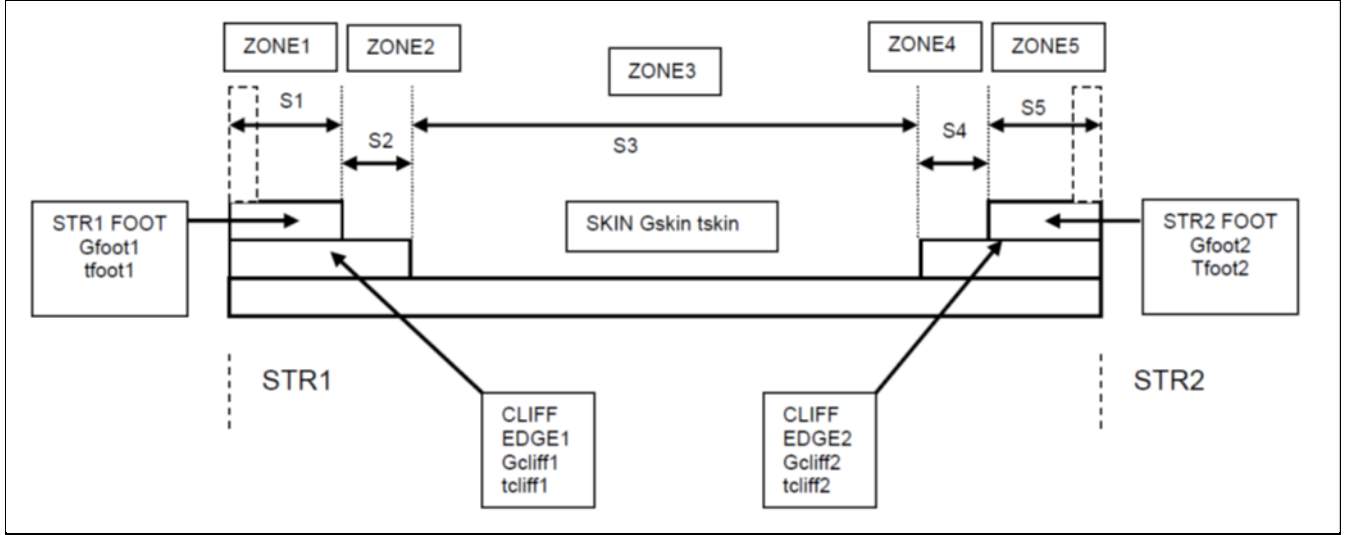


Figure 7. Additional contributors to Torsional Stiffness

According with this method the value that are going to be used to adjust torsional stiffness will be the skin material G Modulus (G_{skin}).

In order to reproduce the torsional behavior, a new $G_{equivalent}$ will be determined and implemented in the GFEM skin shell elements. The new equivalent G Modulus is determined by summing the end rotation (previous Equation) of each component of the Lifting Surface box in Figure 7.

$$\frac{T \cdot L}{4 \cdot A^2 G_{Equivalent}} \oint \frac{ds_{total}}{t_{skin}} = TERM''_1 + TERM''_2 + TERM''_3 + TERM''_4 + TERM''_5$$

Where each $TERM_i$ represents the part of the torsional rotation due to each one of the elements in the figure (skin, stringer foot, cliffedge, ...).

Starting from previous equation, we can derive:

$$G_{Equivalent} = \frac{S_{total}}{TERM_1 + TERM_2 + TERM_3 + TERM_4 + TERM_5}$$

Being

$$TERM_1 = \frac{S_1 \cdot t_{sk}}{Fact_{eff} \cdot G_{STR1} \cdot t_{STR1} + G_{sk} \cdot t_{sk} + G_{CE1} \cdot t_{CE1}}$$

$$TERM_2 = \frac{S_2 \cdot t_{sk}}{G_{sk} \cdot t_{sk} + G_{CE1} \cdot t_{CE1}}$$

$$TERM_3 = \frac{S_3}{G_{sk}}$$

$$TERM_4 = \frac{S_4 \cdot t_{sk}}{G_{sk} \cdot t_{sk} + G_{CE2} \cdot t_{CE2}}$$

$$TERM_5 = \frac{S_5 \cdot t_{sk}}{Fact_{eff} \cdot G_{STR2} \cdot t_{STR2} + G_{sk} \cdot t_{sk} + G_{CE2} \cdot t_{CE2}}$$

The effectiveness of the stringer flange in picking up shear from the skin has to be taken into account. When the stringers, cliff edge or skins are bonded there should be a significant shear carrying capability and this factor [Fact_{eff}] is set at 90% effectiveness. When they are co-cured, this factor is set at 100% effectiveness.

The result is an overall increased torsional stiffness (GJ) in the Lifting Surface box accounting for the stringer flange and cliffedge without increasing the bending stiffness.

For each variation in properties of the skin and stringers, the factor to be applied to the skin G value will be different. Within the FEM, each skin panel thickness is represented with a separate material card (MAT2). This permits the user to precisely modify the G₃₃ value for each skin panel thickness (in red) according to the expression used to determine G_{equivalent}.

PSHELL	*	990000001	199000001	2.57599998E+00	299000001
*		1.00000000E+00	0	1.00000000E+00	0.00000000E+00
*		-1.28799999E+00	1.28799999E+00	0	0
MAT2	*	199000001	7.21501719E+04	2.31117754E+04	-1.77180336E-05
*		5.12229648E+04	-8.97039368E-04	2.43165254E+04	1.58000000E-06
*		0.00000000E+00	0.00000000E+00	0.00000000E+00	2.00000000E+01
*		0.00000000E+00	0.00000000E+00	0.00000000E+00	0.00000000E+00
*		0	0	0	0
MAT2	*	299000001	6.54258711E+04	3.11173301E+04	1.92188635E+03
*		4.19361445E+04	1.92188623E+03	3.23220801E+04	1.58000000E-06
*		0.00000000E+00	0.00000000E+00	0.00000000E+00	2.00000000E+01
*		0.00000000E+00	0.00000000E+00	0.00000000E+00	0.00000000E+00
*		0	0	0	0

Table 6. Modified G33 in MAT2 in NASTRAN

Remark: Using this approach, the **stress** output cannot be extracted directly from GFEM. To get the right Stress output it is necessary to read Axial and Shear flows and divide by the corresponding AE depending the zone.

4.2 Spar Modelling

The spars are structural elements with typical construction of a shear beam. They are composed of a web, two chords (upper and lower) and some stiffeners of the web.

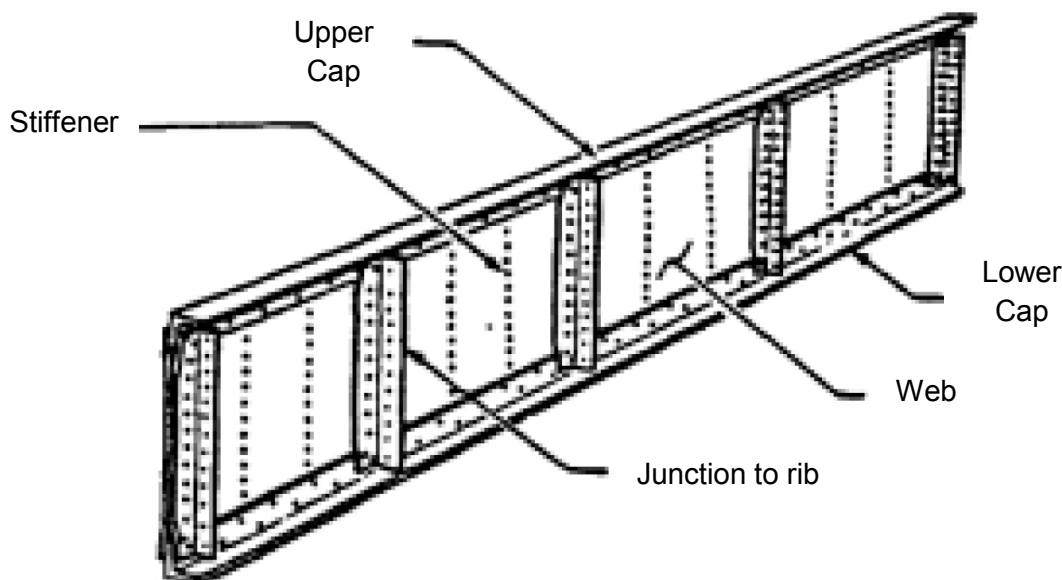


Figure 8. Typical Spar Structural components

The most difficult aspect is to idealize the web because it usually presents a stiffness which is function of the applied load. There are two clearly differentiated types of spar webs:

4.2.1 Spar with Shear resistant web

They have bending stiffness in their plane, axial load and shear. At low loading levels (below the buckling allowable of the web), every web behaves as shear resistant.

Metallic mechanized spars and composite spars. These webs will be idealized with QUAD4 elements.



Figure 9. Loads in a Shear Resistant Spar Web

Spar Modelling

- Web
 - Elements: The skin panel will be idealized with 1 – 3 CQUAD4 elements (depending on the longitudinal stiffeners represented)
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT1 or MAT2: with the thickness of the base skin and with properties according to classic laminate theory.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8.
 - PSHELL/MAT2 Shear Only properties for the web vanes with holes or shear-only capability
- Longitudinal stiffeners
 - Elements: CROD for tension-compression only or CBAR for relevant inertia stiffeners
 - Properties/Material: PROD / PBAR cards with MAT1

- 2 (usually) or 4 longitudinal stiffener can be defined depending on the spar structure definition.
- Transversal Stiffeners
 - Elements: CROD
 - Properties/Material: PROD/MAT1 cards, with A_{tot} equal to the global net area of the vertical stiffeners along the Spar vane (*).

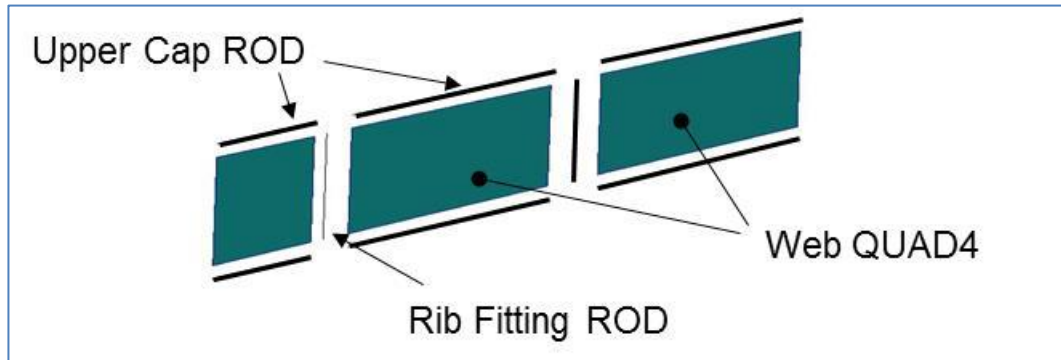


Figure 10. Basic Spar modelling

(*) The vertical stiffeners of the web are usually not idealized, except those located at the position of the rib interface. They work exclusively in tension-compression and will be idealized with CROD elements of area equal to the sum of the areas of the stiffeners (included in the vane) plus the areas of the pads.

4.2.2 Spar with Web in incomplete diagonal tension

In some designs, part of the applied shear is supported by diagonal tension (post-buckle of the web), so that its stiffness depends on the level of applied load. This behavior is clearly non-linear but in the idealization of this type of web, it is usual to assume the web working in a pure state of shear and use the results obtained to make a detailed analysis in diagonal tension. The conventional spars usually have this type of webs.

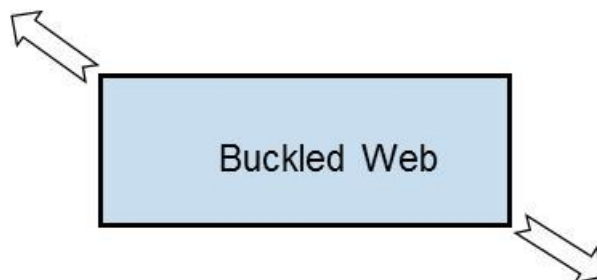


Figure 11. Loads in a Spar Web in incomplete diagonal tension

Spar Modelling

- Web
 - Elements: The skin panel will be idealized with 1 – 3 CQUAD4 elements (depending on the longitudinal stiffeners represented)

- Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2 Shear Only properties for the web vanes with holes or shear-only capability (introducing $t_{\text{equivalent}}$ in PSHELL and only G_{33} in MAT2 and occasionally, for dynamic reasons, it may be necessary in these cases to give small values to the components G_{11} and G_{22} - in a MAT2).
 - PSHELL/MAT1 or MAT2: with the thickness of the base skin and with properties according to classic laminate theory, for membrane+bending panels.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8 for membrane+bending panels.
- Spar Caps and Longitudinal stiffeners
 - Elements: CROD for tension-compression only or CBAR for relevant inertia stiffeners
 - Properties/Material: PROD / PBAR cards with MAT1
 - 2 (usually) or 4 longitudinal stiffener can be defined depending on the spar structure definition.
- Transversal Stiffeners
 - Elements: CROD
 - Properties/Material: PROD/MAT1 cards, with A_{tot} equal to the global net area of the vertical stiffeners along the Spar vane (*).

4.2.3 Spar Caps modelling

If the area associated with the spar cap is not significantly larger than the area of the stringers of the skin, they can be idealized as CROD elements (like the stringers), subsequently correcting the force and outputs of the model by the factor.

$$h/h'$$

Being:

- h : Distance between the outer surfaces of the upper and lower skins (distance of the nodes in the model).
- h' : Distance between the centers of gravity of the upper and lower coverings (panel plus stringers)
- To be consistent with the criteria followed in the idealization of stringers, the separation between the two idealizations (CROD or CBAR/CBEAM) is established for a value of the parameter h/h' of 1.1, so that in values lower than this will be used CROD elements and for higher values CBAR/CBEAM elements (see Figure 12).
- The area associated with each cap $A_{\text{cap}} = A_0 + A_{\text{pad}} + A_{\text{skin}} + A_{\text{eff-web}}$:
 - A_0 of the cap section
 - A_{pad} : the area of the pad-ups
 - A_{skin} : the area of skin above the spar cap and not counted in the skin itself.
 - $A_{\text{eff-web}}$: For conventional spars (shear only web), an additional area is added to the cap equivalent to the effective area in compression of the sheet (approximately 15 times the thickness), $A_{\text{eff-web}} = 15 \cdot t^2$

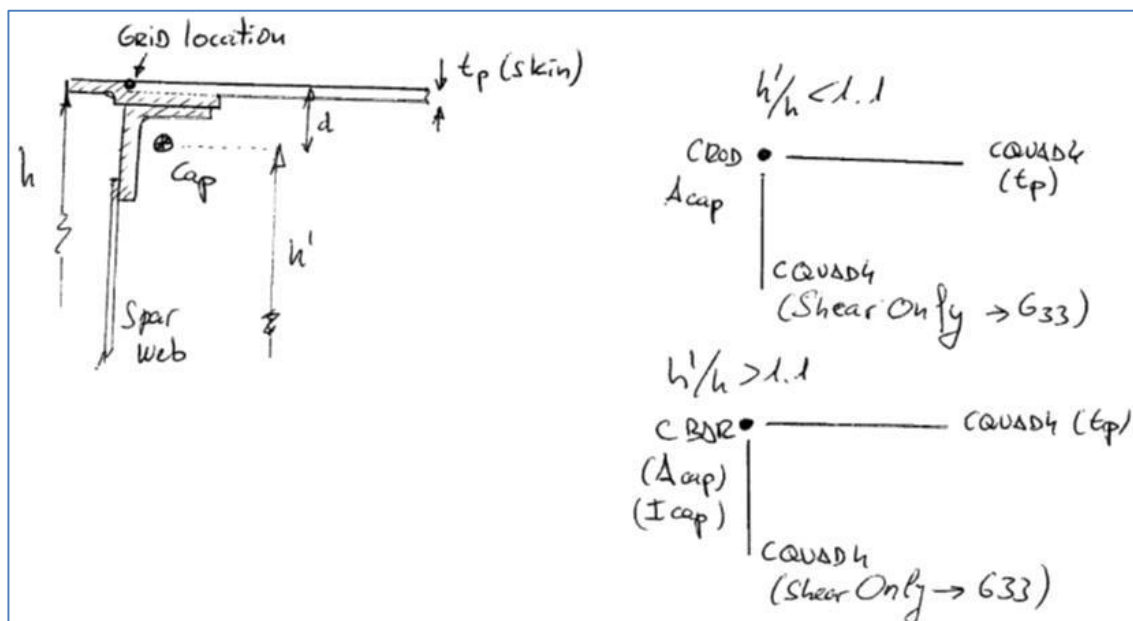


Figure 12. Idealization of the caps with incomplete diagonal tension

4.2.4 Spar lightening holes

If a spar web contains big lightening holes compared with the spar web vane (see Figure 13), it is assumed that the web can take shear only loads

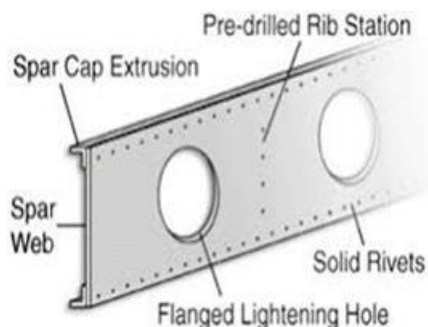
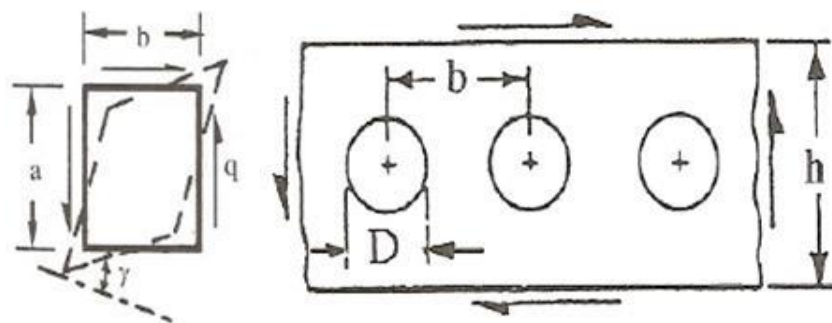


Figure 13. Spar web lightening holes

To determine the equivalent thickness some Detailed FEM can be generated as in Ref. 15.

An approximate solution is to use formulae as the proposed by Ref. 16.



$$t_{eq} = \left(\frac{G \cdot t}{G_{eq}} \right) \cdot \left(1 - \frac{D}{b} \right) \cdot \left(1 - \left(\frac{D}{h} \right)^3 \right)$$

$$\text{if } G = G_{eq} \Leftrightarrow t_{eq} = t \cdot \left(1 - \frac{D}{b} \right) \cdot \left(1 - \left(\frac{D}{h} \right)^3 \right)$$

Figure 14. Equivalent thickness for shear only webs with lightening holes (Ref. 16).

4.2.5 Spar Elements Orientation

According to the following figures, the Spar element orientation is defined as follows:

- Web: $X_{element}$ from root to tip, $Y_{element}$ according to Figure 15 or Figure 16.
- Longitudinal Stiffener: $X_{element}$ from root to tip.
- Transversal stiffener: $X_{element}$ from lower to upper skin.

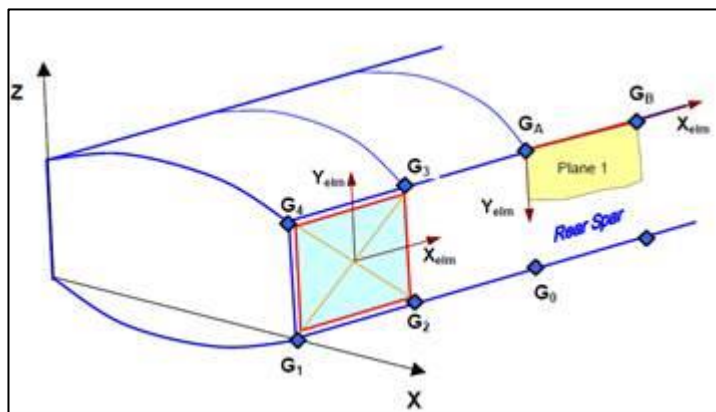


Figure 15. Wing / HTP Spar Elements Axes definition

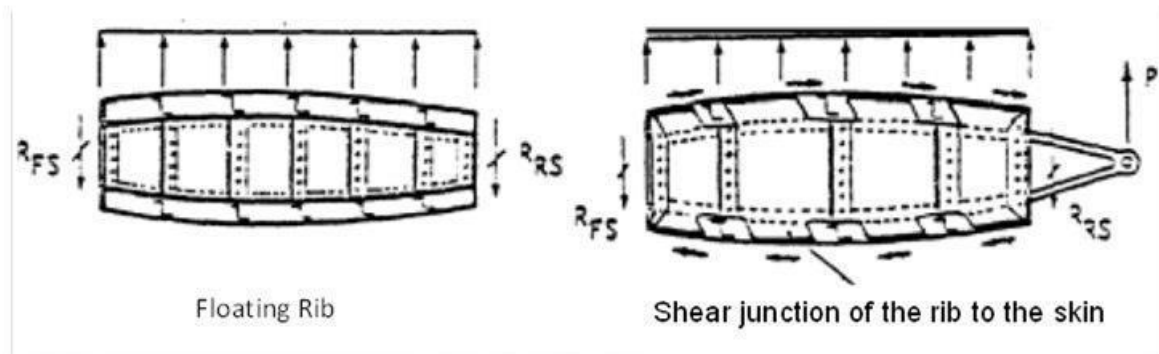


Figure 17. Types of Ribs

General Rib Modelling

- Web
 - Elements: The skin panel will be idealized with 1 (usual) – 3 CQUAD4 elements (depending on the longitudinal stiffeners represented)
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2 Shear Only properties for the web vanes with holes or shear-only capability (introducing $t_{equivalent}$ in PSHELL and only G_{33} in MAT2 and occasionally, for dynamic reasons, it may be necessary in these cases to give small values to the components G_{11} and G_{22} - in a MAT2).
 - PSHELL/MAT1 or MAT2: with the thickness of the base skin and with properties according to classic laminate theory, for membrane only or membrane+bending panels.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8 for membrane+bending panels.
- Rib Cap and Longitudinal stiffeners
 - Similar criteria for the Rib caps can be used than the ones established for the Spar caps
 - Elements: CROD for tension-compression only or CBAR for relevant inertia stiffeners
 - Properties/Material: PROD / PBAR cards with MAT1
 - 2 caps (usually) or 4 (2 caps and 2 longitudinal stiffeners) can be defined depending on the rib structure definition.
- Transversal Stiffeners
 - Elements: CROD
 - Properties/Material: PROD/MAT1 cards, with A_{tot} equal to the global net area of the vertical stiffeners along the Spar vane (according to Figure 18 and example formulae).

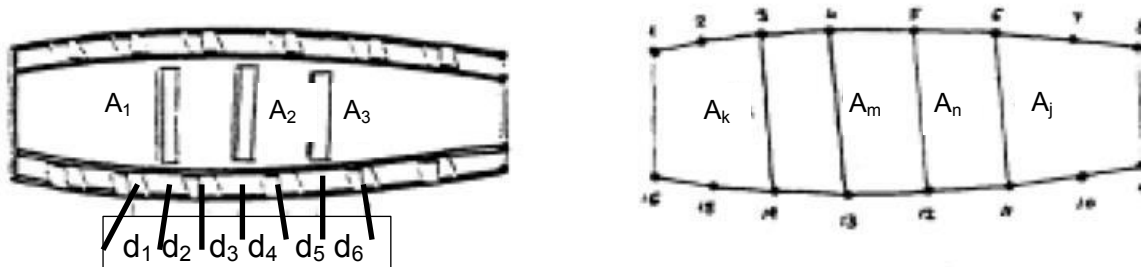


Figure 18. Rib Vertical Stiffeners: Real vs Idealized structure

$$A_k = \frac{A_1 \cdot d_2}{d_1 + d_2}$$

$$A_m = \frac{A_1 \cdot d_1}{d_1 + d_2} + \frac{A_2 \cdot d_4}{d_3 + d_4}$$

$$A_n = \frac{A_2 \cdot d_3}{d_3 + d_4} + \frac{A_3 \cdot d_6}{d_5 + d_6}$$

$$A_j = \frac{A_3 \cdot d_5}{d_5 + d_6}$$

Exceptions

- Special ribs with severe local loads will require an integral construction. The mesh must be modified to preserve the correct load path in the area of the introduction of the load on the rib. This modification will be imposed by the type of idealization that is made of the fittings load introduction.
- In the case of special ribs with severe load input (for example, wing-fuselage ribs or joint of the Engine Mounting System), it will be necessary to make a finer mesh whose greater complexity is compensated by the large variations in the distributions of flows and stresses that withstands that rib. The nodes will be placed preferably at the intersections of the stiffening elements so that the loading path is reproduced as faithfully as possible.

4.3.1 Rib Lightening holes

Similar approach can be followed as explained in 4.2.4.

4.3.2 Rib Elements Orientation

According to the following figures, the Rib element orientation is defined as follows:

- Web: X_{element} from Rear Spar to Front Spar, Y_{element} according to Figure 19 or Figure 20.
- Longitudinal Stiffener: X_{element} from Rear Spar to Front Spar.
- Transversal stiffener: X_{element} from lower to upper skin.

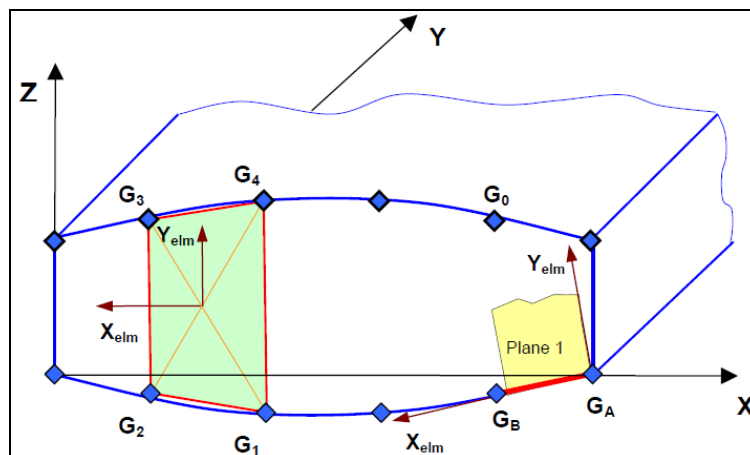


Figure 19. Wing/HTP rib element axes and global axes

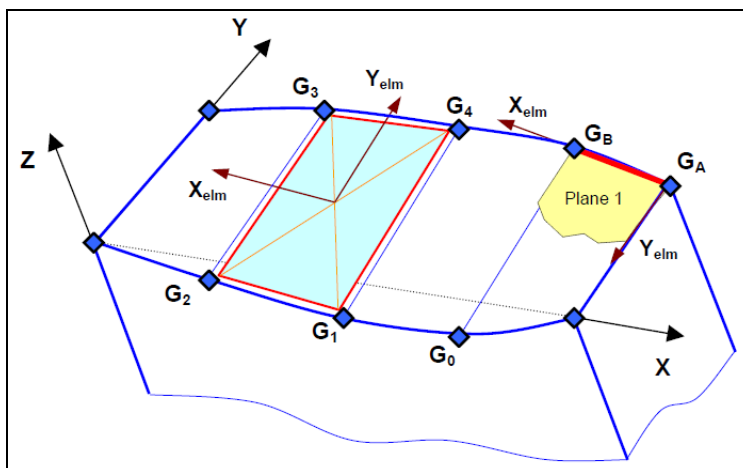


Figure 20. VTP rib element axes and global axes

5. FUSELAGE MODELLING

In the idealization of a typical fuselage structure for a GFEM the following hypotheses are made:

- The nodes will be located on the outer surface of the skin at the points of intersection of the plane of the frames with the stringers, unless for some reason (as presence of an opening to be idealized or load entry at an intermediate point) should be added intermediate nodes.
- The panels of the skin support the torsion and shear loads applied to the fuselage through shear and the bending loads in tension-compression.
- The compression skin is not fully effective when entering in a buckling state.
- Small openings should be ignored in this model and only large ones, such as windows, passenger doors or landing gear should be represented by their incidence in the internal load redistribution.
- The frames work exclusively on their plane.
- The stringers support the bending of the fuselage when subjected to tension-compression. They do not contribute to the torsional stiffness of the fuselage and, except in extremely necessary cases, its bending stiffness is neglected.
- Some Doors are designed not to transfer load to the fuselage (except pressure loading) and may not be included in the GFEM. The doors that induces continuity to the frames (cargo doors for example) must be idealized to give continuity to the circumferential loads.

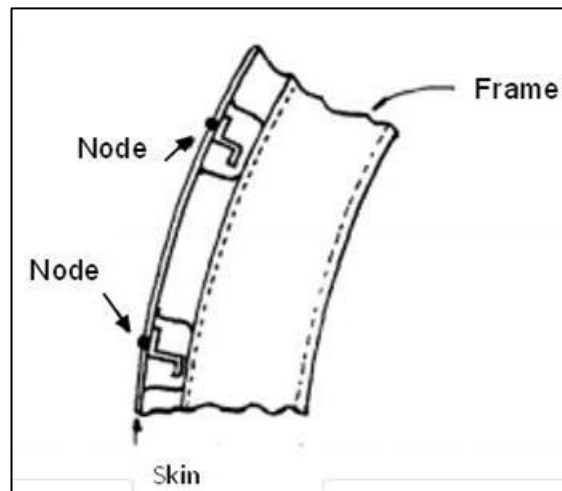


Figure 21. Typical Fuselage Frame

5.1 Fuselage Skin

Fuselage Skin modelling, as in the case of lifting surfaces, it depends on the Manufacturing process and on the relative inertia of the stringers and Frames related to external radius of the Fuselage (R/R' , according to Figure 22).

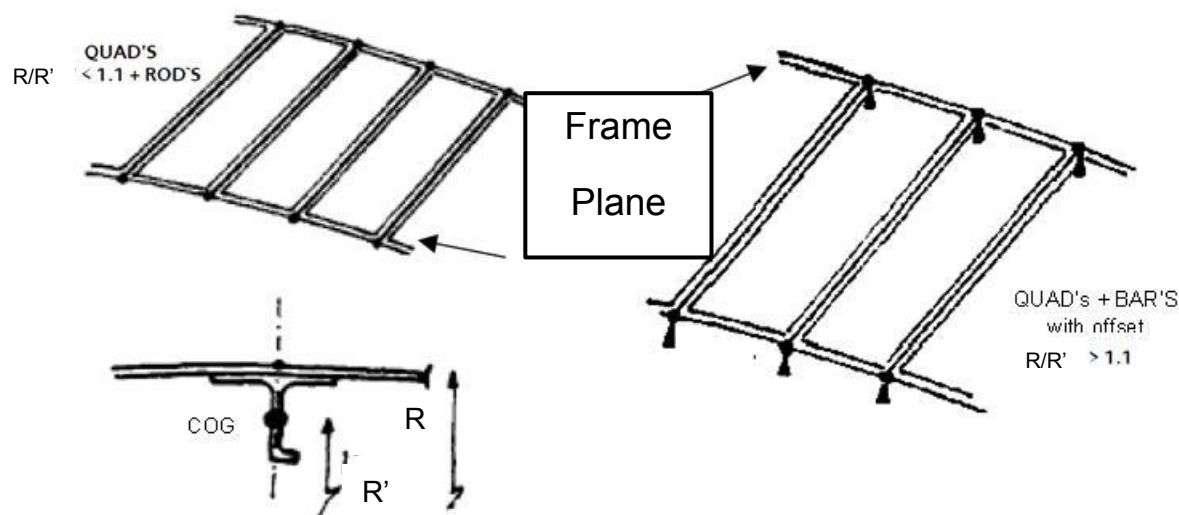


Figure 22. Basic Fuselage Skin Modelling

5.1.1 Metallic Skin Modelling ($R/R' < 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties: PSHELL card with the thickness of the base skin and with membrane properties only (MID1, without bending capability- MID2=0).
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD elements
 - Properties: PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists.
- Material: MAT1 card with E modulus.
- Element Orientation:
 - Shell: $X_{element}$ parallel to Fuselage axis, to the end of fuselage, $Z_{element}$ looking outwards, $Y_{element}$ from top to bottom (RHS) and in opposite direction in LHS. See attached Figure.

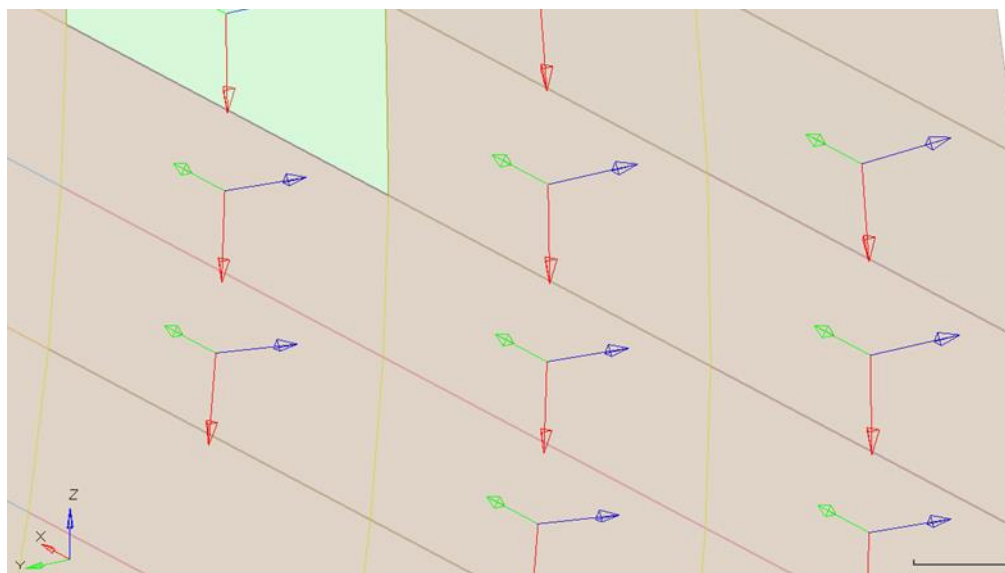


Figure 23. Fuselage Skin Element Orientation

- Stringer: X_{element} from nose to tail.

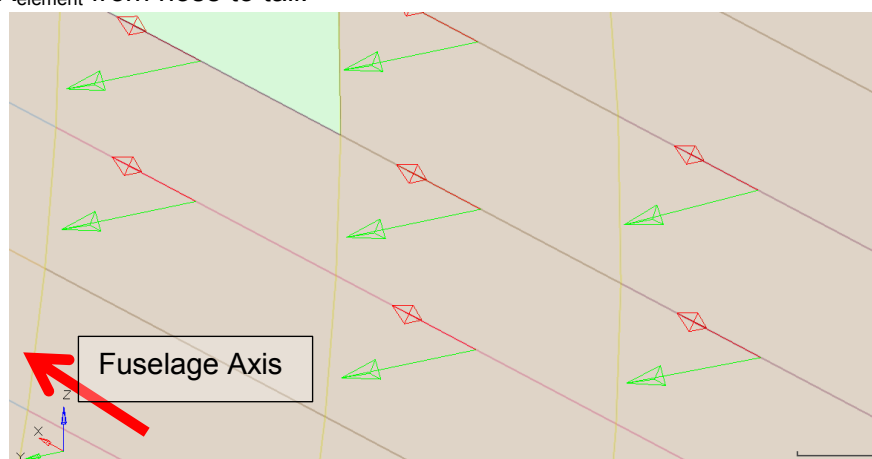


Figure 24. Fuselage Stringer Elements Orientation

5.1.2 Metallic Skin Modelling ($R/R' > 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties: PSHELL card with the thickness of the base skin and with membrane properties only (MID1, without bending capability- MID2=0).
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD or CBAR elements
 - Properties:
 - PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists factorized with coefficient h/h' .

- PBAR plus offset values, with the area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists, and corresponding Inertias.
- Material: MAT1 card with E modulus.
- Element Orientation:
 - Shell: X_{element} parallel to Fuselage axis, to the end of fuselage, Z_{element} looking outwards, Y_{element} from top to bottom (RHS) and in opposite direction in LHS.. See Figure.
 - Stringer: X_{element} from nose to tail.

5.1.3 Solid Laminate Composite Skin Modelling ($R/R' < 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2: with the thickness of the base skin and with properties according to classic laminate theory.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8.
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD elements
 - Properties/Material: PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists. MAT1 corresponding to apparent elastic modulus of the stringer section ($\Sigma E_x A / \Sigma A$), according to Classical Laminate Theory..
- Element Orientation:
 - Shell: X_{element} parallel to Fuselage axis, to the end of fuselage, Z_{element} looking outwards, Y_{element} from top to bottom (RHS) and in opposite direction in LHS.. See Figure.
 - Stringer: X_{element} from nose to tail.

5.1.4 Solid Laminate Composite Skin Modelling ($R/R' > 1.1$)

- SKIN:
 - Elements: The skin panel will be idealized with CQUAD4 elements
 - Properties/Material: PSHELL or PCOMP cards
 - PSHELL/MAT2: with the thickness of the base skin and with properties according to classic laminate theory.
 - PCOMP/MAT8: PCOMP card with real stacking sequence and MAT8.
- STRINGERS:
 - Elements: The stringers or stiffeners will be idealized with CROD or CBAR elements
 - Properties/Material:
 - PROD card without torsion properties and with an area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists factorized with coefficient h/h' .
 - PBAR plus offset values, with the area A equal to the stringer cross section plus the area of extra thickness ('pad'), if exists, and corresponding Inertias.
 - MAT1 corresponding to apparent elastic modulus of the stringer section ($\Sigma E_x A / \Sigma A$), according to Classical Laminate Theory.
- Element Orientation:
 - Shell: X_{element} parallel to Fuselage axis, to the end of fuselage, Z_{element} looking outwards, Y_{element} from top to bottom (RHS) and in opposite direction in LHS.. See Figure.
 - Stringer: X_{element} from nose to tail.

5.1.5 Exceptions

- Inter-Frame Nodes: When the model has nodes between frames, it is necessary to use elements with stiffness out of the plane. In these cases it will be used CBAR elements (CBEAM for non-linear or tapered section) without offset and with the appropriate bending stiffness (the corresponding to the stringer plus the area of extra thickness) instead of CROD elements. For thick skins and for obtaining more accuracy, the stiffness of the panel can be added to the stiffness of the CBAR element relative to its midline (adding the bending stiffness in the CQUAD4 element).
- Stringers with relevant inertia: When the stiffeners have an appreciable height regarding to the height of the torsion box (for example, they differ by an order of magnitude only) it is necessary to take into account explicitly the effect of the position of the center of gravity of the resistant area. If not, while the value of the internal force or stresses can be corrected later to take this effect into account (as described below), the error committed in the stiffness would be relative high and not easily correctable. In this case CBAR elements with offset will be used, being this the distance from the center of gravity of the stringer plus 'pad' to the reference line of the nodes. For non-linear studies, CBEAM elements without offsets will be used instead of CBARs with offsets, with the bending properties corrected to the outer point of the skin (reference line of the nodes), Ref. 18.
- When the two previous situations occur simultaneously it will be used CBAR elements with offset (linear analysis) or CBEAM without offset (non-linear analysis) and corrected inertias.

5.2 Frames

Typical Fuselage Frames Geometry in Transport A/C are described in Figure 25. Depending on the level of detail used in the model and the type of postprocessing tool, the Frame can be modelled in GFEM by means of 1D or 2D elements as shown in Figure 26 and next sections.

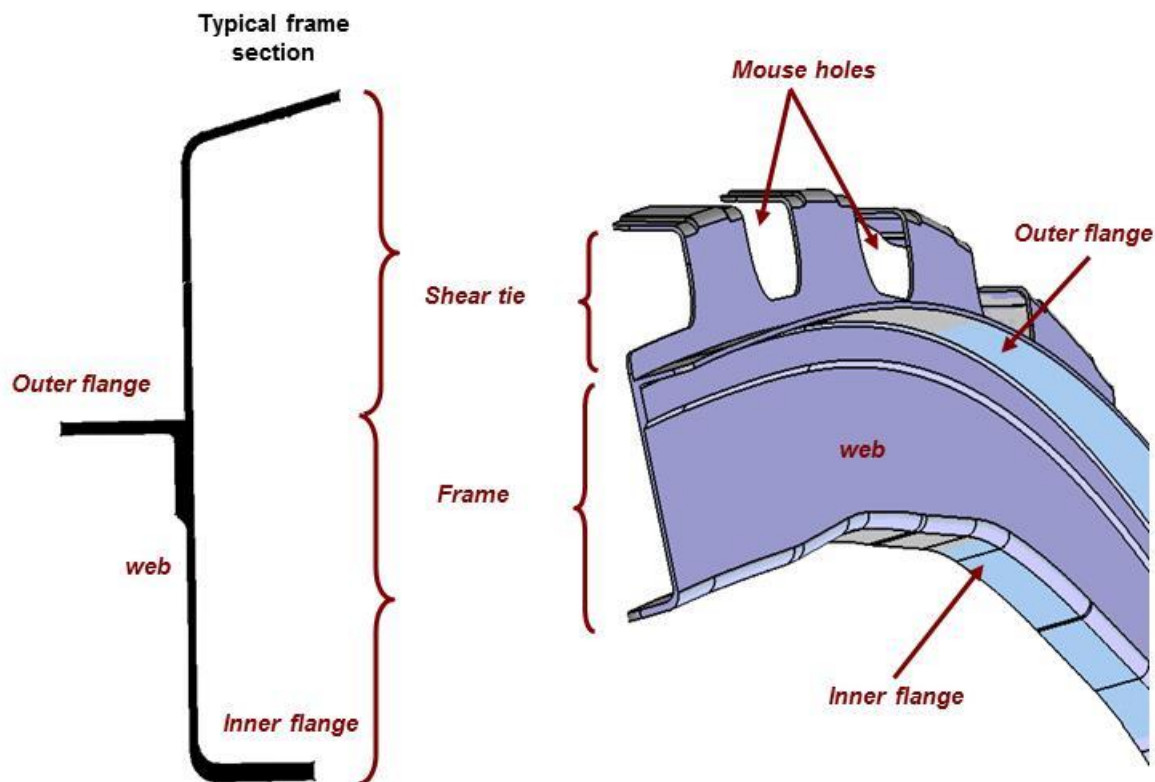


Figure 25. Typical Fuselage Frame Geometry

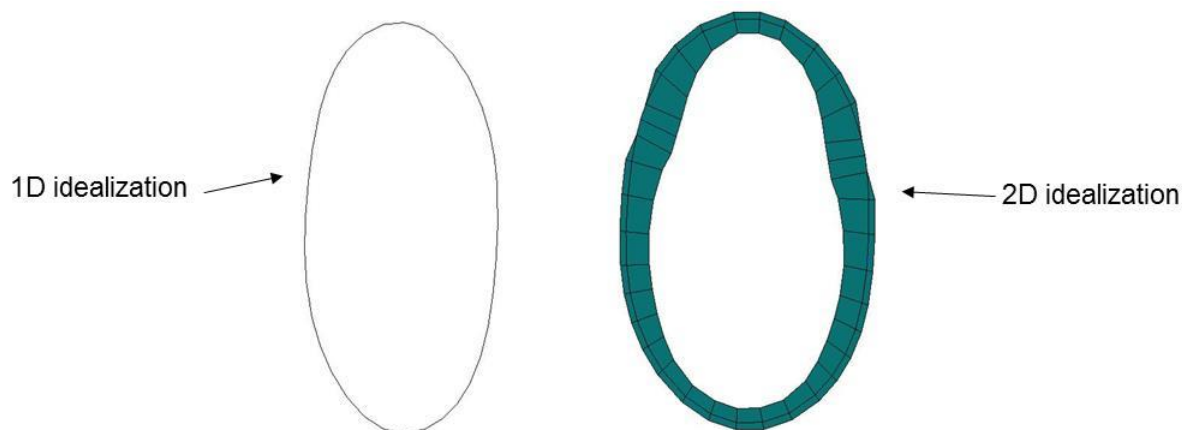


Figure 26. Frame Modelization

5.2.1 1D Frames Modelling

The frames will be idealized with CBAR elements of area equal to the area of the cross section of the frame (excluding the clips or 'babettes' that don't have continuity to let the stringers continuity) and with offset, being this the distance from the centre gravity of the normal section of the frame to the position of the node (note that this offset is given by default in the coordinate system of displacements of the node (analysis) and that the component outside the plane of the frame will be zero). These CBAR elements will only have properties in the plane of the frame, that is, the moment of inertia of bending in the plane of the frame and the effective area for shearing in that same plane, neglecting the moment of inertia regarding to the perpendicular plane, the product of inertia if it exists, the torsional stiffness and the effective area for shearing normal to the plane of the frame.

In cases of pressurization load (this type of load is introduced as a radial load on the frames), the proposed idealization of frames and skin leads to an oversized frame because the contribution of the skin to withstand this type of load is neglected. Analyses made show that the skin can carry out up to 80% of the pressurization loads, thus unloading the frames.

In these pressurization cases, the load that must be supported by non-idealized doors/windows must be transmitted to the structure as force and moment, so it may be necessary locally in an area of the frame to assign bending properties outside the frame plane and even torsion to the CBAR elements. These internal forces will be transmitted to surrounding area. The use of pin-flag in these elements should be avoided to reduce dynamic problems.

Some Airbus DS Legacy tools require that frames are represented by CBAR and taking into account that this program correctly interprets the forces in the CBAR elements that idealize the frames, it is necessary to take special careful on selecting as axis systems of these CBAR elements the axis Ye (+) (or Ze (+), depending the definition defined) so that it points towards the interior of the fuselage. This axis, along with the X axis, (in the tangential direction of the frame), will define the plane of the frame.

The properties of the CBAR elements can be introduced by means of the PBARL cards defined in Ref. 1.

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
PBARL	PID	MID	GROUP	TYPE	-	-	-	-	
	DIM1	DIM2	DIM3	DIM4	DIM5	DIM6	DIM7	DIM8	
	DIM9	etc.	NSM						

Table 7. Typical PBARL definition

In addition to the standard library of sections generated by MSC (GROUP = MSCBML0), libraries of sections generated ad-hoc can be used, such as the existing library of sections in Airbus DS (GROUP = LOCBMLS1, only for Nastran v70.5), which contains sections such as the 'C-variable' and the 'G-variable'. These beam libraries are managed in accordance with what is described in Ref. 16.

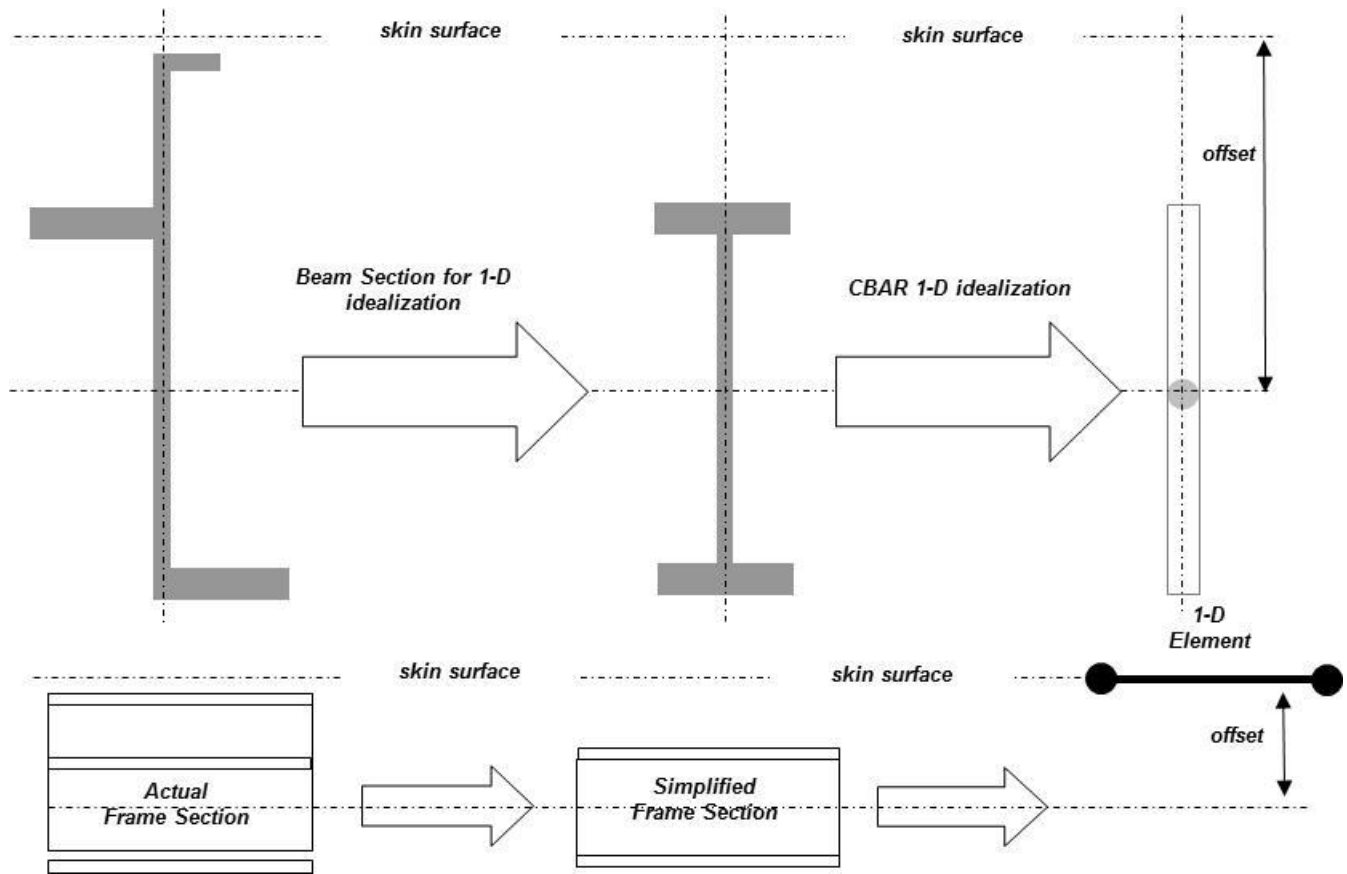


Figure 27. 1D Frame Idealization

- **Frame Element Orientation:** X_{element} from top to bottom, Y_{element} looking inwards, Z_{element} from nose to tail (RHS) and in opposite direction in LHS. See Figure.

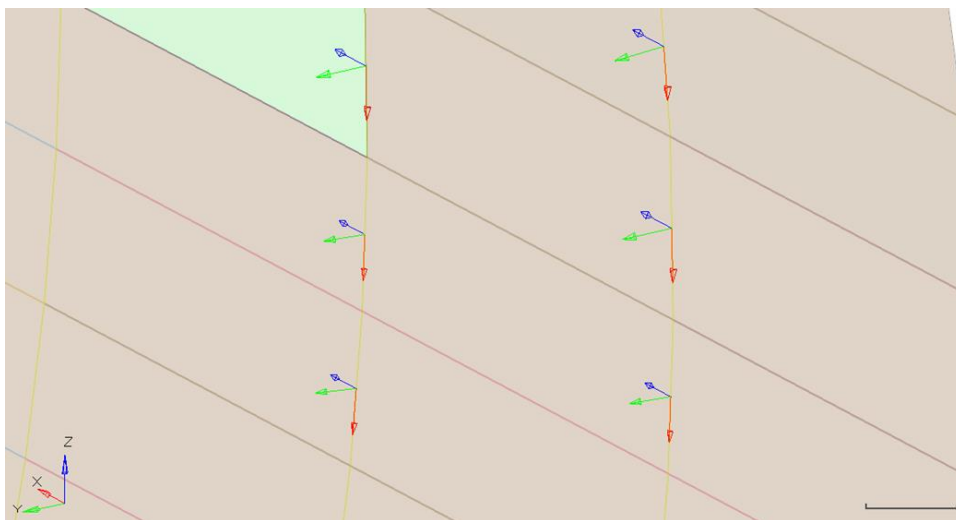


Figure 28. Fuselage Frame Element Orientation

5.2.2 2D Frames Modelling

For small diameter fuselages and large frames, the height of the frame section can be significant compared to the radius of the fuselage. In this case, it will be necessary to carry out a finer idealization with the help of nodes placed in the centre of gravity of the interior cap of the frame, as shown in Figure 18. The caps of the frame will be ROD elements with its own area and the web will be a QUAD4 element with membrane properties exclusively. For the idealization (if exists) of the clips or 'babettes' (shear attachment between frame-skin) it is necessary to keep in mind that in the model this union is continuous, whereas in the real structure it is discontinuous due to the passage of the stringers. Therefore, the thickness of the element QUAD4 (G_{33} and occasionally, for dynamic reasons, it may be necessary in these cases to give small values to the components G_{11} and G_{22}) will model the stiffness of the junction according with the expression shown in section 6.3

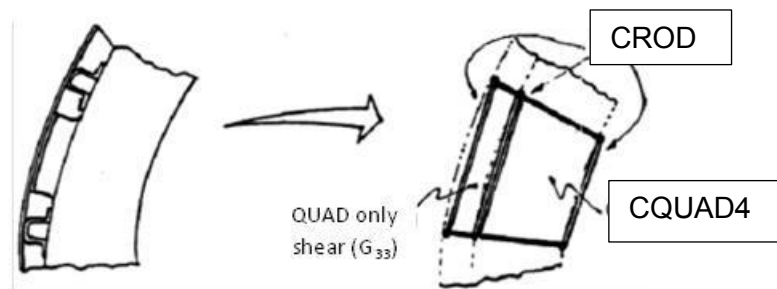


Figure 29. Typical 2D Frame modelling

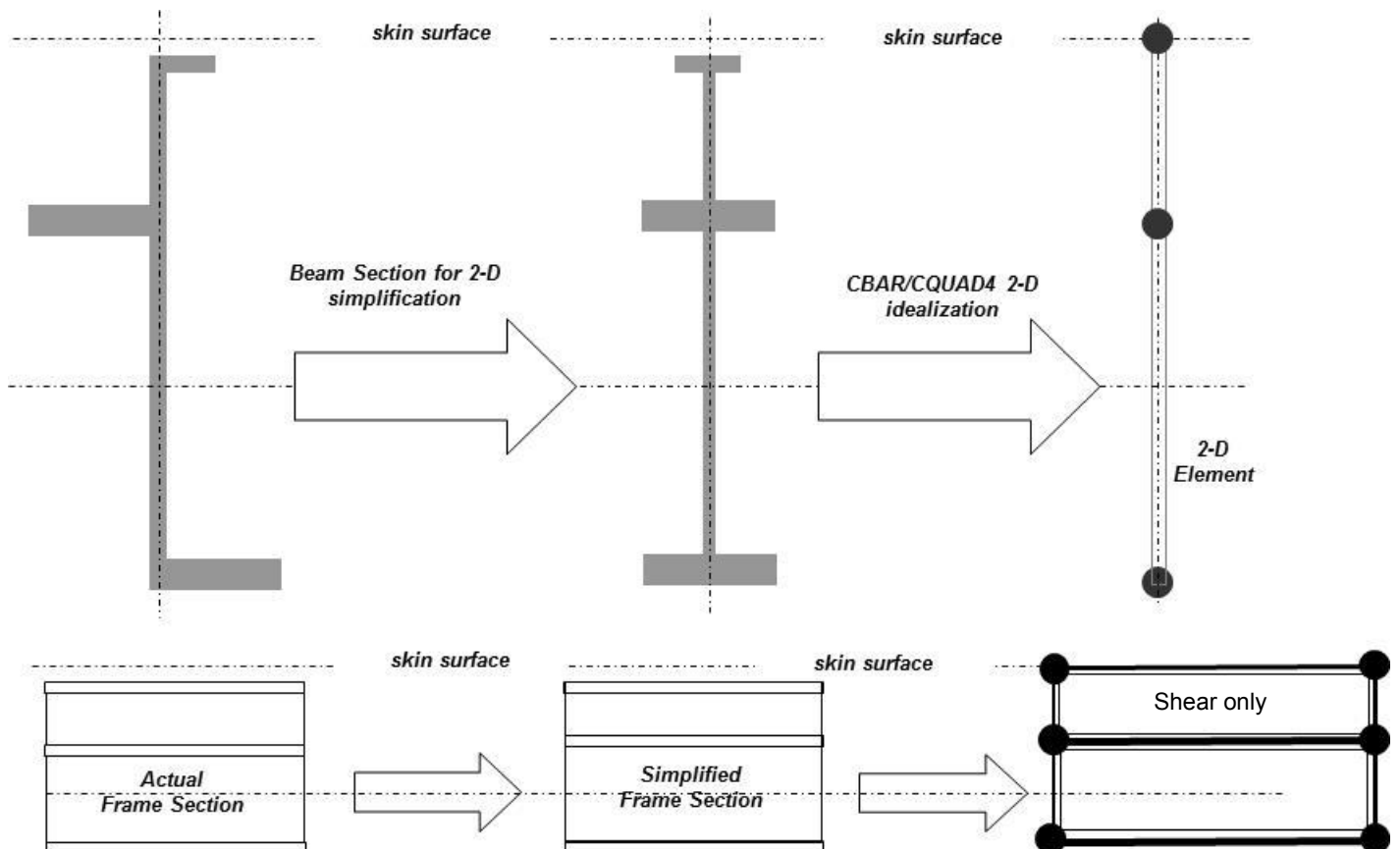


Figure 30. 2D Frame Idealization

Element orientations according to criteria defined for 1D Frames definition. See attached Figures

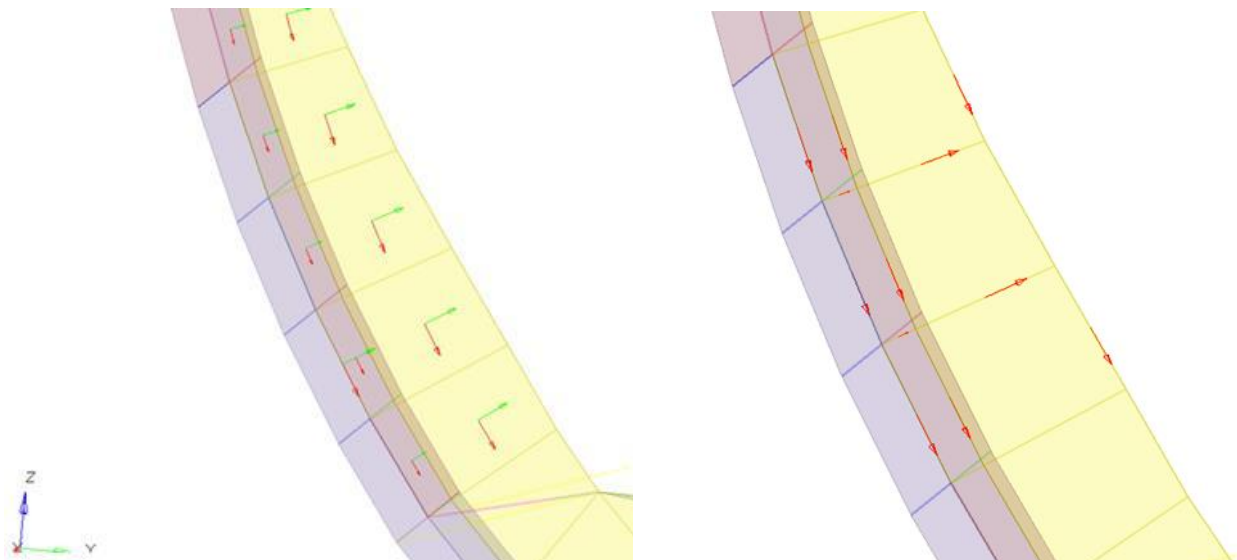


Figure 31. 2D Frame Elements Orientation

5.3 Fuselage Windows

Windows are usually not idealized specifically, but simulated by

- some QUAD4 element which thickness “t” reproduces the same stiffness as the real panel with the opening trying to reproduce exactly the main way of work of the element, usually shear-only,
- and the difference in tension-compression stiffness regarding to the previous element, if significant, will be introduced with ROD elements, in radial and axial direction.

Usually a DFEM of the window is performed in order to obtain the $t_{\text{equivalent}}$ and A_{axial} and A_{radial} .

As an alternative, a basic modelling of the window can be introduced in Fuselage GFEM.



Figure 32. Simplified Window representation (Right) and detailed (Left)

5.4 Floors

The floors in passenger or cargo cabin are supported on a structure formed by the following elements (see Figure 33):

- Beams in the longitudinal direction of the airplane, located in the positions where the rails are placed, which act as support for the floor and attachment of the seats and other elements (galley, cabinets...). This beam is considered a truss type with vertical and diagonal elements between the frames (or it can also be a beam web-cap with a shear web).
- Transverse crossbeams corresponding to the frames, on which the floor is supported and which are joined to the previous mentioned beams

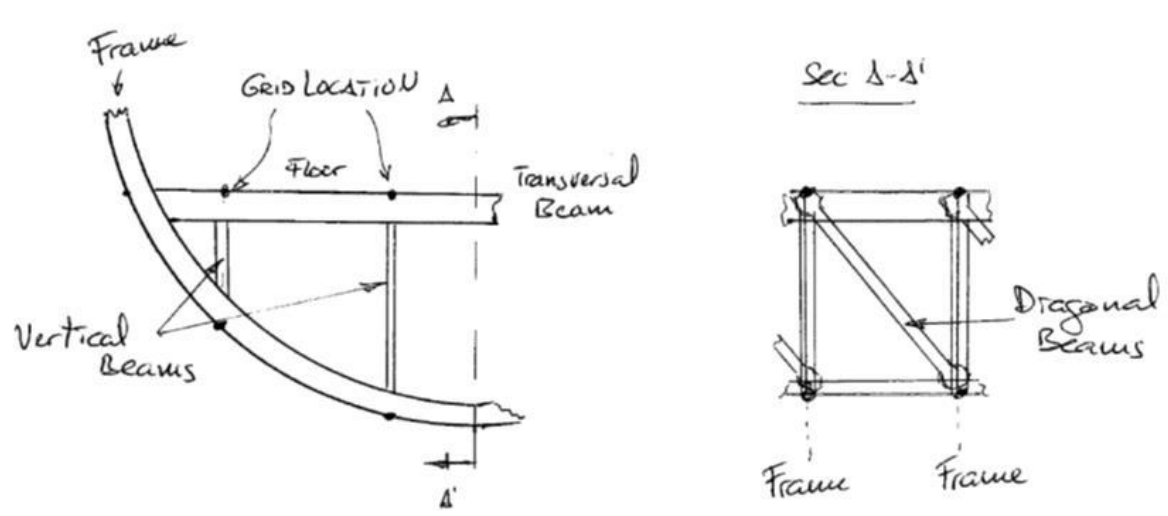


Figure 33. Floor structural components

In the model, the nodes will generally be located at the intersection outer points of the structural elements because these points are more direct and easy of calculate, but it would also be possible to locate them in the centres of gravity of the elements in the same previous intersections. Since the main purpose of these elements as floor supports is to transmit the load coming from them to the rest of the structure, and to be applied mainly through the rails, to make possible a homogeneous distribution of the load between the frames and the longitudinal beam, the elements corresponding to the crossbars must be idealized as CBAR elements with offset (or without them when the nodes are in the centre of gravity of the elements, these CBARs will be replaced by CBEAM without offsets for non-linear studies) and with flexural properties in the plane perpendicular to the floor that contains the element, so that the applied load can be redistributed in shear and bending between the frame and the longitudinal beam.

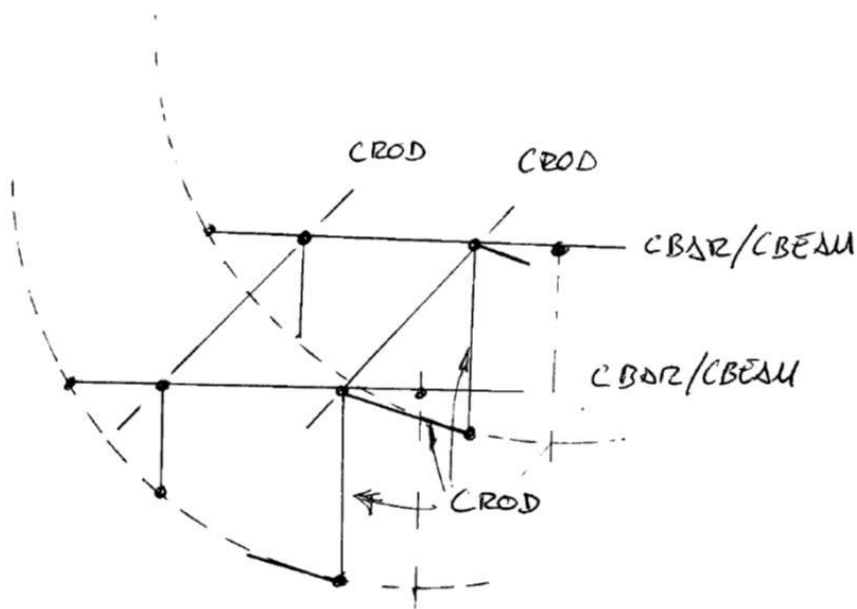


Figure 34. Typical Floor Elements Modelling

The remaining elements like the upper cap of the longitudinal beam and the vertical and diagonal bars of the longitudinal beam will be idealized depending on the capacity or not of the floors to carry out bending loads of the airplane. In the event that they do not have this capacity, both the upper cap of the longitudinal beam and the vertical and diagonal bars must only support tensile or compressive loads, so they will be idealized as CROD elements without taking into account the possibility of torsion. Floor panels will not be idealized. In case they contribute to support primary loads of the aircraft, the upper cap of the longitudinal beam will be idealized with CBAR elements with offset (or without them when the nodes are at the centre of gravity of the elements. These CBARs will be replaced by CBEAM without offsets for nonlinear studies) and with flexural properties in the plane perpendicular to the floor that contains the element. The vertical and diagonal posts (if they exist) will be idealized as CROD elements without taking into account the possibility of torsion. The floor panels will be idealized with CQUAD4 elements (usually they are sandwich, chapter 10).

Floor support structures with web-cap type will be idealized by means of CROD elements for the caps and CQUAD4 elements (G_{33} and occasionally, for dynamic reasons, it may be necessary in these cases to give small values to the components G_{11} and G_{22}) only with shear stiffness for the web, using CBAR elements (CBEAM for non-linear) only where the effects of bending of the cap are very significant.

5.5 Bulkhead

A bulkhead is a type of closed frame that usually is end of fuselage pressurized area or fuel tank boundary.

These types of components are modelled by means of CQUAD4 / CTRIA3 with shell properties (PCOMP or PSHELL with membrane and bending properties) and CBAR elements for bulkhead stiffener, with offset when necessary.

6. FITTINGS MODELLING

The fittings are structural elements with the function of connect and join other structural elements together. Its size and way of working is very diverse, being practically impossible to systematize it. But, in any case, its idealization within a general model, as is the reason for this instruction, should not be very detailed, since the interest does not lie in analyzing the fitting itself but the general structure and load path distribution.

Therefore, the idealization of the fitting will be made through a simplification of its way of working, to ensure the following two essential points:

- The stiffness of the simplified model of a fitting must be equal to the stiffness of the real fitting.
- The simplified model must respect the load paths of the real fitting and introduce the load in the structure at the same points.

The idealization of the fitting will be as simplified as possible taking into account the two previous mentioned items. Thus, in cases of complicated fittings of great stiffness compared with the adjacent structure, it is enough to use an RBE2 element that connects the node where the load is applied to the fitting (it will be considered independent in its six degrees of freedom) with the nodes of the structure where the fitting is connected (see section 10.2.4). If the stiffness cannot be considered as infinite, an idealization will be carried out that complies with the previous points being the most used elements CROD, CBAR/CBEAM and CTRIA3/CQUAD4, as seen in the classification that follows.

It must always be taken into account that the model can be singular for lateral loads (loads outside the plane of the fitting). The basic solution to this problem is to connect the lateral DOF with a rigid element to the contiguous rib or frame.

All the interfaces between structural zones (wing-fuselage, pylon-wing, hinge axis between flap and wing, etc.) should be represented by mean of duplicated nodes, thus creating an interface in which some nodes belong to a part of the structure (for example, the wing torsion box) and the others, for example, flaps (coincident in location). Once this is done, the duplicated points are joined by means of CBUSH/CELASx elements. These CBUSH or CELASx will be used to read the Interface forces between components.

CBUSH is the preferable solution because it is an element with corresponding Element Coordinate System in which the element forces are expressed.

CELASx is a Scalar Element and has not an element system. The load depends on the DOF connected in the element. Many legacies GFEM used CELASx for interfaces.

CELASx elements should always be connected to the same criterion due to the sign of the load at the output, for example, a criterion would be calling "GRID1" to the external point, flap, and "GRID2" to the inner, torsion box). The stiffness of these CELASx elements will correspond to the bolted joint stiffness connection, for example.

The CBUSH/CELASx stiffness are usually several orders of magnitude higher than that of the contiguous structural elements, so that they can be considered rigid, however, a bad conditioning of the stiffness matrix of the structure must be avoided so, as indicative stiffness, the following values can be used:

- Wing-Fuselage Interface $\rightarrow K=10^8$ N/mm
- Flap-Wing Interface $\rightarrow K=10^6$ N/mm

Figure 35 is a good example of this idealization

In the following subsections will be proposed modelling rules for some types of fittings that cover the most common cases in aeronautical structures, without forgetting the possibility of using the RBE2 elements described above. These types of fittings are:

6.1 “V” Fittings

The “V” Fittings are the typical type of fittings used to connect movable surfaces, as Ailerons or elevators.

There are different types of “V” fittings:

- With web
- Without web

Typical fitting representation is the one shown in Figure.

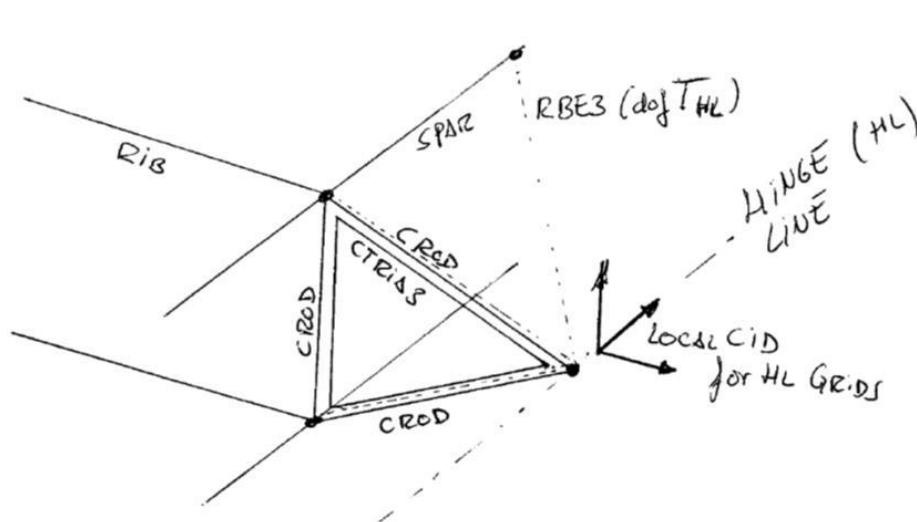


Figure 35. Typical ‘V’ Fitting modelling

The GRID located at the Hinge line will be defined with Hinge Line Co-ordinate System as Analysis CID. According to chapter 8, multiple GRIDs can be defined at the same location in order to capture interface loads and represent Fail-Safe scenarios and movable surface rotation.

The basic elements used will be:

- 1D Elements (CROD or CBAR) for Fitting Flanges, if exist.
- CTRIA3 shell element for Fitting web (if exists)

The Element properties will be calculated in order to have equivalent stiffness in both directions perpendicular to Hinge Line to the real fitting.

6.2 Special Fittings

Many type of fittings can be modelled in a similar way than proposed in previous section using CTRIA3 + CROD/CBAR, with different combination of Properties to take into account the case of Shear Fittings, Tension Fittings, or the cases of Fitting with lateral bending capabilities.

In all the cases the approach should be the same: to perform a DFEM of the fitting and create a simplified model with the same stiffness in the relevant DOFs.

For more complex fittings, as shown in Figure 36, a more detailed representation can be done of the fitting at GFEM level, to represent the real load path.

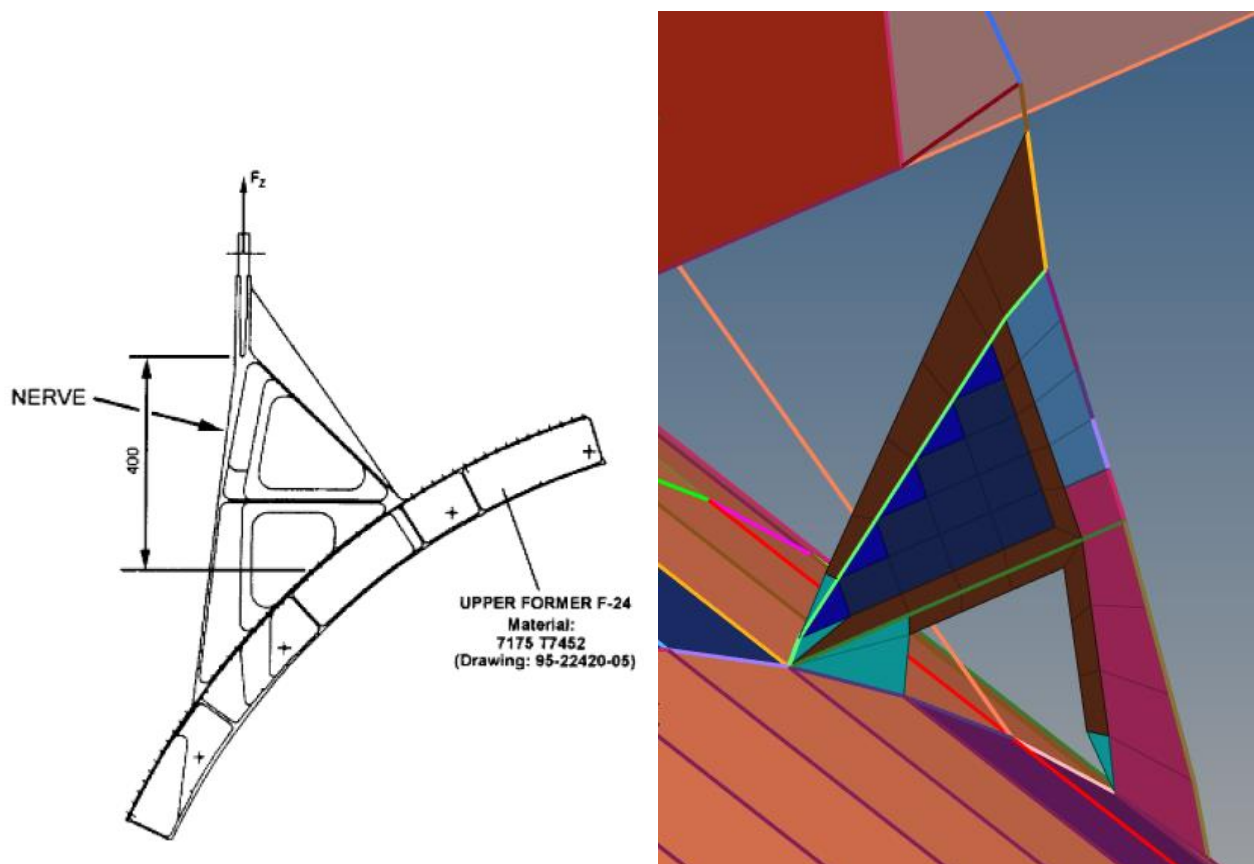


Figure 36. C295 Fuselage to Wing Fitting Drawing and Model

7. SPECIAL INTERFACES MODELLING

7.1 Fittings Fail-Safe conditions Modelling

Most relevant A/C structural Interfaces can be modelled to represent Fail-Safe scenarios using GFEM linear static analysis.

Many of the interfaces of structural components are defined as Fail-Safe. In that cases each one of the fittings can fail and the rest of the structure can sustain the re-distributed load.

These types of events can be modeled by using MPC and MPCADD cards.

- At each interface a set of MPC's are defined (linking the DOF affected by the interface) in serial with the corresponding CBUSH/CELASx elements that will be used to read the interface loads.
- MPCADD for intact structure condition can be build adding all MPC's that links every fitting active at intact scenario.
- A specific MPCADD can be generated to represent each Fitting failure, by calling all fitting MPC's except the ones affecting the failed fitting
- A specific SUBCASEs set can be built to analyze the requested load cases with each failed scenario

An example of this modelling is shown in next tables. Table 8 shows a fail-safe case control, where a unique load case can be analyzed with different MPC conditions representing different failure scenarios. Table 9 shows different MPCADD corresponding intact and failure scenarios: MPCADD will content all MPC's linking non-failed interfaces. Finally, Table 10 shows the MPC cards at every interface point. As shown in scheme of Figure 37, an additional intermediate GRID point is need to be added at the same position of the interface.

\$*****	
\$	Failure Upper Fwd Fitting RHS HTP-Fuselage
\$*****	
\$	
SUBCASE 121023	=== → Different SUBCASE ID
SUBT= A04029 (F00) :	Fail Up Fwd Fit RHS HTP-Fuselage
LOAD= 100023	=== → Same Load Condition 100023
MPC = 4001	=== → MPCADD Failure 1
\$	
\$*****	
\$	Failure Center Fwd Fitting RHS HTP-Fuselage
\$*****	
\$	
SUBCASE 122023	=== → Different SUBCASE ID
SUBT= A04029 (F00) :	Fail Center Fwd Fit RHS HTP-Fuselage
LOAD= 100023	=== → Same Load Condition 100023
MPC = 4002	=== → MPCADD Failure 2
\$	
\$*****	
\$	Failure Lower Fwd Fitting RHS HTP-Fuselage
\$*****	
\$	
SUBCASE 123023	=== → Different SUBCASE ID
SUBT= A04029 (F00) :	Fail Lower Fwd Fit RHS HTP-Fuselage
LOAD= 100023	=== → Same Load Condition 100023
MPC = 4003	=== → MPCADD Failure 3
\$	

Table 8. Case Control for Fail-Safe Scenario

\$#####								
\$ MPC Intact Structure								
\$#####								
MPCADD	1000	100	10	11	12	13	14	15
16	17	18	19	20	21		49011	49012
49023	49031	49032	59011	59012	59023	59031	59032	
45121	45123	45241	45243	55121	55123	30022	30023	
55241	55243	69032	69011	69013	69021	69023	65121	
65122	65211	65212	65301	65302	12151	12152	30021	
12153	12161	12162	12163	12453	12463	20811	20812	
20813	20991	20993	21171	21173	30811	30812	30813	
30991	30993	31171	31173	20021	20022	20023	45051	
45052	45053	45001	45002	45003	45005	65002	65003	
65006	65051	65052	65053	65001	25810	25990		
\$#####								
\$ MPC Fail-safe Cases								
\$#####								
\$ MPC Right HTP-Fuselage Interface.....								
\$ HTP Front Upper Fitting Failure								
MPCADD	4001	100	10	11	12	13	14	15
16	17	18	19	20	21		30022	30023
49023	49031	49032	59011	59012	59023	59031	59032	
45121	45123	45241	45243	55121	55123	20023	30021	
55241	55243	69032	69011	69013	69021	69023	65121	
65122	65211	65212	65301	65302	12151	12152	20022	
12153	12161	12162	12163	12453	12463	20811	20812	
20813	20991	20993	21171	21173	30811	30812	30813	
30991	30993	31171	31173	20021	49022	45051	25990	
45052	45053	45001	45002	45003	45005	65002	65003	
65006	65051	65052	65053	65001	25810			
\$								
\$ HTP Front Center Fitting Failure								
MPCADD	4002	100	10	11	12	13	14	15
16	17	18	19	20	21		49011	49012
49031	49032	59011	59012	59023	59031	59032	30023	
45121	45123	45241	45243	55121	55123	20023	30021	
55241	55243	69032	69011	69013	69021	69023	65121	
65122	65211	65212	65301	65302	12151	12152	30022	
12153	12161	12162	12163	12453	12463	20811	20812	
20813	20991	20993	21171	21173	30811	30812	30813	
30991	30993	31171	31173	20021	20022	49013	49033	
45052	45053	45001	45002	45003	45005	65002	65003	
65006	65051	65052	65053	65001	25810	25990	45051	
\$								
\$ HTP Front Lower Fitting Failure								
MPCADD	4003	100	10	11	12	13	14	15
16	17	18	19	20	21		49011	49012
49023	30022	30023	59011	59012	59023	59031	59032	
45121	45123	45241	45243	55121	55123	20023	30021	
55241	55243	69032	69011	69013	69021	69023	65121	
65122	65211	65212	65301	65302	12151	12152	20022	
12153	12161	12162	12163	12453	12463	20811	20812	
20813	20991	20993	21171	21173	30811	30812	30813	
30991	30993	31171	31173	20021	49022	45051	25990	
45052	45053	45001	45002	45003	45005	65002	65003	
65006	65051	65052	65053	65001	25810			
\$								

Table 9. MPCADD Cards for Fail-safe Analysis

\$#####							
\$ MPC Links HTP/FUSELAGE (RHS)							
\$ Intermediate GRID (to define MPC)							
GRID	1803304	0	21674.7	1465.0	1620.5	0	
GRID	1803305	0	21668.5	1465.0	1521.3	0	
GRID	1803306	0	21674.7	1465.0	1422.0	0	
\$ HTP Front Upper Fitting connection							
MPC	49011	1803304	1	-1.0	103304	1	1.0
MPC	49012	1803304	2	-1.0	103304	2	1.0
MPC	49013	1803304	3	-1.0	103304	3	1.0
\$ HTP Front Center Fitting connection							
MPC	49022	1803305	2	-1.0	103305	2	1.0
MPC	49023	1803305	3	-1.0	103305	3	1.0
\$ HTP Front Lower Fitting connection							
MPC	49031	1803306	1	-1.0	103306	1	1.0
MPC	49032	1803306	2	-1.0	103306	2	1.0
MPC	49033	1803306	3	-1.0	103306	3	1.0
\$							
CELAS2	1871380	1.0E+8	1803304	1	490001	1	
CELAS2	1872380	1.0E+8	1803304	2	490001	2	
CELAS2	1873380	1.0E+8	1803304	3	490001	3	
\$							
CELAS2	1871381	1.0E+8	1803305	1	490002	1	
CELAS2	1872381	1.0E+8	1803305	2	490002	2	
CELAS2	1873381	1.0E+8	1803305	3	490002	3	
\$							
CELAS2	1871382	1.0E+8	1803306	1	490003	1	
CELAS2	1872382	1.0E+8	1803306	2	490003	2	
CELAS2	1873382	1.0E+8	1803306	3	490003	3	
\$							
CELAS2	1872383	1.0E+8	103307	2	490004	2	
CELAS2	1873383	1.0E+8	103307	3	490004	3	
CELAS2	1872384	1.0E+8	103308	2	490005	2	
CELAS2	1873384	1.0E+8	103308	3	490005	3	
\$							

Table 10. MPC Cards definition for Fittings Connection

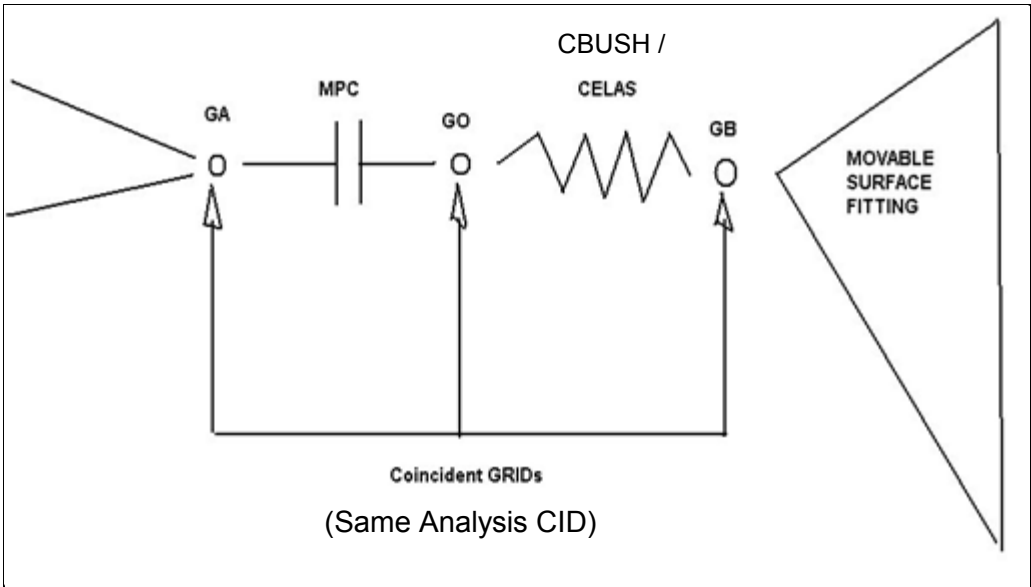


Figure 37. Interface Modelling for Fail-Safe Analysis

7.2 Rotation of Movable surfaces

In this section it will be described some methods of reproducing different rotations of the movable surfaces, obtaining correctly the loads and the stiffness from the point of view of the torsion box with a single model.

7.2.1 Method1: Using Co-ordinate Systems & MPCs

Starting from a typical interface construction as defined in previous paragraph at each interface point along the Hinge line:

- GRIDs GA, GB and G0 are defined with the same Analysis Co-ordinate System, which is the Hinge line CID (HL₀).
- Hinge Line (HL) CID is usually defined with:
 - o origin located at the intersection of the inner fitting with HL
 - o HL axis will be the Y axis for Ailerons or elevators, or Z axis in the case of Rudder
 - o X axis will be perpendicular to the HL pointing rearwards

This configuration will represent the interface with No-rotation of the movable surface.

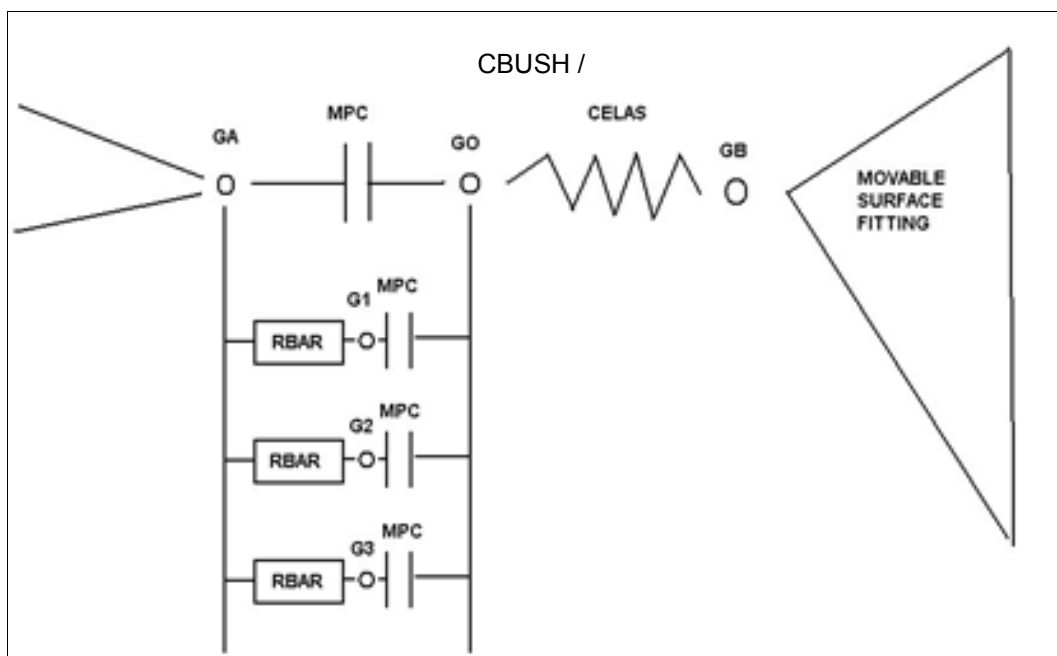


Figure 38. Interface modelling for Movable Surfaces Rotation

To generate the rest of rotations of the movable surfaces that is needed to represent:

1. One additional CID will be created for each rotation analyzed HL_i (where i= δ_{rot}):
 - i. The new CID HL_i will have the same HL axis
 - ii. The rest of the axes will be the result of applying a rotation around the HL axis according the surface deflection represented (see Fig. XX)
 - iii. For example, $\delta_{rot} = -30^\circ, -20^\circ, -10^\circ, 10^\circ, 20^\circ, 30^\circ$,

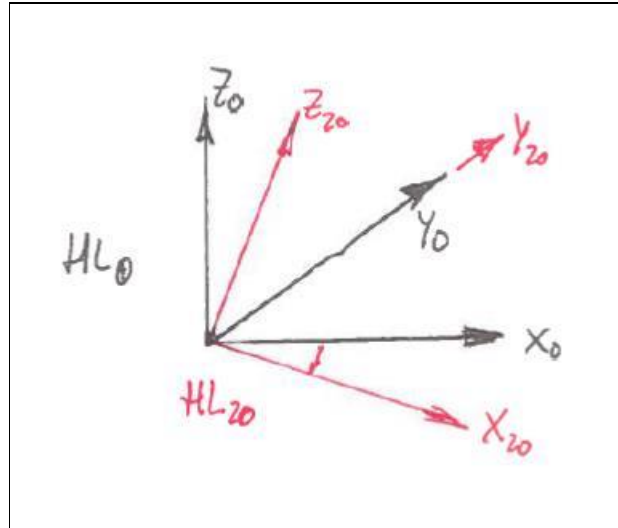


Figure 39. Rotated CID around HL0

2. One additional GRID point G_i will be created for each rotation at each fitting position.
 - i. This new GRID will be located at the same position than G_A , G_B and G_0
 - ii. The analysis CID will be the corresponding HL_i CID.
3. In order to connect the new GRIDS to the interfaces, a set of RBAR/RBE2 and MPC will be needed, As shown in Figure 38.
 - i. If G_A is dependent in the MPC created for 0 deg. rotation (as in Figure 38), the RBAR will link G_A and G_i
 - ii. If G_0 is dependent in the MPC created for 0 deg. rotation, the RBAR will link G_0 and G_i
 - iii. RBAR will link all 6 DOF
 - iv. One MPC must be created for each DOF affected by interfaces (at least 3 translational dofs) linking each G_i dof with G_0 dof
 - v. MPC ID will be selected with care in order to make easy to generate MPCADD cards
4. A specific MPCADD will be created for each rotation, calling for the MPC's generated for one specific rotation and not the ones for intact/no rotation.

For the actuator associated with the movable surface a similar technique is used for the MPCs, now using a cylindrical coordinate system (CORDSC) to define the position of the interface point, and modifying the area of the actuator (CROD/CBAR) according to the length that the bar has in reality for each deflection, to reproduce the real stiffness of the set ($K=A \cdot E/L$) since, as in the case of the fittings, the surface is never geometrically deflected, but only the loads are rotated mathematically. A different actuator will be connected, corresponding to each surface deflection.

7.2.2 Method2: Using DOFs projections & MPCs

An alternative method for simulating movable surfaces rotations can be done using no news CID or RBARs, but MPC's that rotates the loads according to the deflections and its projection:

$$u_h^{box} = u_h^{elev} \cos \delta + w_h^{elev} \sin \delta$$

$$w_h^{box} = -u_h^{elev} \sin \delta + w_h^{elev} \cos \delta$$

For a deflection of $\delta=15^\circ$, the corresponding MPC will be as follows:

```
MPC      59812115 5912400 1 -1.00000 5912200 1 0.96593
          5912200 3 -0.25882
MPC      59812315 5912400 3 -1.00000 5912200 1 0.25882
          5912200 3 0.96593
```

In this case, the MPC involved actuator interface will contain the projection of the DOFs according the local angle of rotation of the actuator.

7.2.3 Method3: Physical Rotation

Depending on the Project requirements, it can be necessary to represent physically the rotation of the movable surfaces.

This is not the preferable solution because the additional management cost in terms of files and output files.

Using this method a different model will be generated for each rotation. And a different output file should be analyzed for each rotation.

In order to minimize the updating error, when using this approach, a unique Nastran model (bdf or dat file) will be created to represent movable surface nodes and elements) and a different interface file with rotated Co-ordinate System (CID) and interface nodes and elements will be used for each rotation. Using this approach the updating process of the model of the movable surface is simplified.

8. LOADS AND BOUNDARY CONDITIONS

One of the critical points for GFEM modelling is the way to introduce the loads in the model and how to constraint it.

There are different types of loads that can be applied to the model and there are different types of representation of the same load case.

8.1 Pressure Loads

Constant Pressure loads as Fuselage Internal Pressure load are usually represented by PLOADx cards and are applied to shell elements located in the pressurized area, as Fuselage skin and bulkheads.

PLOADx usually applies pressure according to element normal. So the pressure value shall be positive or negative according to the normal selected for each group of elements.

For the holes or gaps in the pressurized area we can use:

- PLOAD cards, applying pressure to the edge nodes of the hole or gap
- FORCE card with the net force on the affected are, connected to the structural boundary by means of RBE3

The resultant forces of a pressure load case should be near to zero or very small.

8.2 Fight & Ground Load Cases

In this paragraph we consider every load case resultant from a maneuver, gust, landing or handling condition.

8.2.1 SMT Loads Data

Preliminary Loads information are usually provided by means of a simplified format.

- Some Monitoring Stations (MS) are defined along A/C
- At each MS an integrated value of loads: F_x , F_y , F_z , M_x , M_y , M_z are provided (Shear, Moment and Torque, SMT)
- From these integrated distribution of forces and Moments, the corresponding applied forces and moments can be obtained at each MS. The integration of these applied forces/moments must correspond to the provided information at each Monitoring Station.
- Using RBE3 these applied forces and moments at each MS can be distributed to the GFEM structural nodes

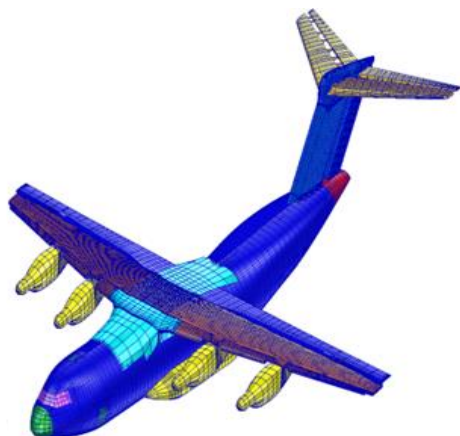
8.2.2 Nodal Loads

The information provided by Loads department in Nodal Loads format is the more accurate to represent A/C loading.

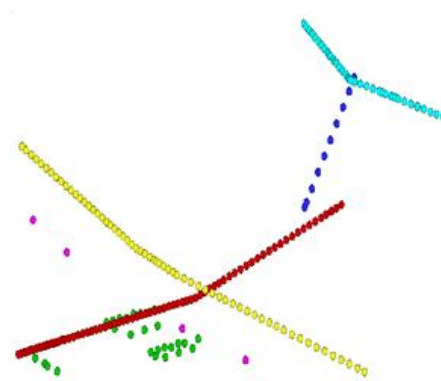
The load cases are equilibrated, so the reactions at the boundary conditions should be marginal.

To generate Nodal Loads, Load department need to have a Loads Interface Model (LIM). LIM is a NASTRAN format Model that contains the A/C wet surface and Inertial points:

- Aerodynamic LIM: NASTRAN Model build with CQUAD4 and CTRIA3 that covers all aerodynamic surfaces. No properties are assigned to the elements. It is coincident to the GFEM in most of the parts. No internal structure is included, but fairings and leading and trailing edge surfaces are included
- Inertial LIM: include all the points that represents the masses of the A/C



LIM: AERO Surfaces



LIM: INERTIA Nodes

Figure 40. LIM: Load Interface Model

8.3 Thermal load Cases

Thermal conditions are applied by means of thermal mapping for different conditions (Hot day, cold day or different temperature scenarios).

These thermal maps are applied to GFEM by means of TEMPD cards applied to GFEM Grid points.

8.4 Combined Load Cases

Different set of combined load cases can be generated combining:

- Fuselage Pressure
- Flight and Ground Load cases
- Thermal Load cases

The combination can be done by means of:

1. SUBCASE creation with the combination of LOAD Cards for Mechanical Load cases and TEMP(LOAD) for thermal load cases
2. Or SUBCOM creation, for linear combination of Mechanical SUBCASEs with Thermal SUBCASEs

8.5 Mass representation

In order to perform Dynamic analysis or Design Optimization, the mass of the structure shall be well represented.

8.5.1 Dynamic Analysis:

- Structural and non-structural masses (systems, fuel) shall be represented by means of CONM2 cards.

- Each CONM2 card will represent the A/C mass corresponding to a section previously established
- Different CONM2 distributions will be generated for different fuel masses.
- Different Fuselage operational or optional items will be also provided by means of CONM2 cards

8.5.2 Design Optimization Analysis

- Structural mass is represented by means of densities in the Material cards in GFEM elements.
- Additional structural mass not taken into account in GFEM (fairings, riveting, secondary structure) shall be represented by means of CONM2 cards.
- Non-structural masses (systems, fuel) shall be represented by means of CONM2 cards.

8.6 GFEM Boundary Conditions

GFEM is a MSC.Nastran model that will be used for Linear Static Analysis. Due to that, the GFEM must be constrained even in the case that the model under the load case is in equilibrium.

Some possible Boundary conditions can be used:

- For a Global A/C GFEM loaded with correlated or equilibrated load cases: Six and only six non co-linear DOF must be selected in order to constraint the model. The reaction at the SPCs should be very small, but must be revised.
 - o e.g. choosing constraints, that residual forces/moments will be minimal or that overall rigid body displacements will be minimized
- For a Global A/C GFEM loaded with not equilibrated load cases: Select an isostatic constraint far from the area that will be analyzed.
- For isolated Component GFEM the constraints should represent the interface DOFs.

9. STRUCTURE STIFFNESS ASSESSMENT

The bending (EI) and torsion (GJ) stiffness of a lifting surface or fuselage along the span can be obtained from the general finite element model, making small modifications in it, through the following procedure.

9.1 GFEM Model Preparation

1. **Elastic Axis:** Define an axis along the span that will be considered as an elastic axis. Although that axis is theoretically the position of the shear centres of each section, it is usual to use for simplicity as an elastic axis the one that links the geometric centres of the ribs. The stiffness will be calculated regarding to this axis and it is necessary to highlight that it will have the breaks that the structure presents.

If we know the location of the shear center or we can obtain the location, we will use this shear center as the reference for this analysis

In many cases, the reference axis for the applied loads is defined in the same way and therefore the elastic axis and reference axis of the loads will coincide. Otherwise, it will be necessary to define a new local coordinate system that represents the elastic axis using the corresponding cards (CORD2R,...). If the structure has an arrow (break), as mentioned above it will be necessary to define an additional local coordinate system for the torsion box.

2. **Reference GRIDs:** Define a GRID node at the intersection of the elastic axis with the plane of each rib. The displacements and rotations of these nodes in the local system of coordinates that represents the elastic axis will allow calculating the bending and torsion stiffness of the element. Therefore, although these points can be defined in any reference system (basic or local), the displacement output will be requested in the local system defined in the previous step (using field 7 of the node definition card).
3. **RBE3:** Define a rigid element RBE3 in each rib position, which only connects the four nodes of each rib in the position of the stringers of the spars with the new node created, as shown in Figure 88. The six dependent degrees of freedom are assigned to this new node. With this configuration it is achieved that the displacement of the central node of the rib is an average of the displacements of the corner nodes. The displacement and rotation of the central node will indicate the displacement and rotation that the rib has. RBE3 card defined will be as following:

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE3	EID		NEWGRID	123456	1.	123	G1	G2	OABC
+ABC	G3	G4							

Where NEWGRID is the Reference GRID and G1, G2, G3 and G4 the identities of the nodes of the rib plane (independent) located at interface of the spars with corresponding rib.

If we use lifting surface root as boundary condition, it is necessary to fix the corresponding NEWGRID point with an SPC card and therefore NEWGRID DOFs must be independent. In order to do that, the RBE3 must be built as follows:

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE3	EID		NEWGRID	123456	1.	123	G1	G2	OABC
+ABC	G3	G4							OABD
+ABD	UM	G1	123	G2	23	G3	2		

Table 11. RBE3 definition for stiffness calculation

4. LOADs definition: Create at the tip of the elastic axis corresponding MOMENT cards to be used for stiffness calculation applied in corresponding Elastic Axis CID
 - Wing / HTP Tip: Mx for Bending, My for Torsion, Mz for lateral bending
 - VTP: Mx for bending, My for Lateral bending and Mz for Torsion
 - Fuselage: Mx for Torsion, My for bending and Mz for lateral bending
5. Rationale: The bending stiffness is derived from the relative rotations produced between two sections when a moment is applied in a direction normal to the elastic axis and contained in the plane of the span applied in the last section of the lifting surface (Mx, in Figure Figure 88). The torsional stiffness is derived from the rotations produced in the sections for a moment in the direction of the elastic axis applied in the last section of the lifting surface (My in Figure Figure 88).

For lifting surfaces that present a change of sweep (kink) at the interface between central torsion box and lateral torsion box will be necessary, as previously mentioned for the definition of the elastic axis, to define two other load cases as the previous ones, referred to the corresponding elastic axis and now applied at a point outside the central zone, area whose stiffness is desired to obtain (three or four ribs away from the interface).

The boundary conditions of the model will not change regarding to the general model. From the output of displacements of the central nodes of the ribs in elastic local axes, the calculation of the stiffness is immediate. In the local axes represented in Figure 88.

9.2 Bending Stiffness Curve

- For wing / Tail Plane bending stiffness, El_x : From the beam bending theory:

$$\frac{M_x}{El_x} = \frac{1}{R}$$

$$R = \frac{dl}{d\theta_x}$$

Discretizing, between sections i and j

$$R = \frac{L_{ij}}{\theta_{x_j} - \theta_{x_i}}$$

$$El_{x_{ij}} = M_x \frac{L_{ij}}{\theta_{x_j} - \theta_{x_i}}$$

Where:

- | | |
|----------------|--|
| $El_{x_{ij}}$ | Is the average bending stiffness between sections i and j. |
| L_{ij} | Is the distance between sections i and j. |
| θ_{x_i} | Is the rotation in the X direction of the central node of section i. |
| θ_{x_j} | Is the rotation in the X direction of the central node of section j. |

Correction: The fact of placing the nodes of the model on the outer surface of the skin and due to the idealization proposed by means of ROD elements (see chapter 4) makes the bending stiffness of the model slightly higher than the stiffness of the real structure. Therefore, the results obtained from the model must be corrected multiplying by the factor:

$$K=(h/h')^2$$

Being:

- h: Distance between the outer surfaces of the upper and lower skins (distance of the nodes in the model).
- h': Distance between the centres of gravity of the upper and lower coverings (panel plus stringers)

With the previous calculation, a constant value factor has been assumed along the span. Otherwise, an averaged value will be used.

- For Fuselage bending stiffness, EI_y : From the beam bending theory:

$$\frac{M_y}{EI_y} = \frac{1}{R}$$

$$R = \frac{dl}{d\theta_y}$$

Discretizing, between sections i and j

$$R = \frac{L_{ij}}{\theta_{y_j} - \theta_{y_i}}$$

$$EI_{y_{ij}} = M_y \frac{L_{ij}}{\theta_{y_j} - \theta_{y_i}}$$

Where:

$EI_{y_{ij}}$	Is the average bending stiffness between sections i and j.
L_{ij}	Is the distance between sections i and j.
θ_{y_i}	Is the rotation in the Y direction of the central node of section i.
θ_{y_j}	Is the rotation in the Y direction of the central node of section j.

9.3 Torsional stiffness

From the torsion theory, the rotation per unit length produced by a Torsion moment M_y (for Wing or HTP) on a beam of stiffness GJ is given by:

$$\frac{d\theta_y}{dl} = \frac{M_y}{GJ}$$

Discretizing:

$$\frac{\theta_{y_j} - \theta_{y_i}}{L_{ij}} = \frac{M_y}{GJ_{ij}}$$

$$GJ_{ij} = M_y \frac{L_{ij}}{\theta_{y_j} - \theta_{y_i}}$$

Where:

GJ_{ij} Is the average torsional stiffness between sections i and j .

L_{ij} Is the distance between sections i and j .

θ_{yi} Is the rotation in the direction and the central node of section i .

θ_{yj} Is the rotation in the direction and of the central node of section j .

The torsional stiffness obtained by means of FEM are usually underestimating the real Torsional stiffness. Some elements are not taking into account as Leading edge or trailing edge. Additionnally, according to 4.17, some corrections should be made for co-bonded or co-cured skin to correct torsional stiffness.

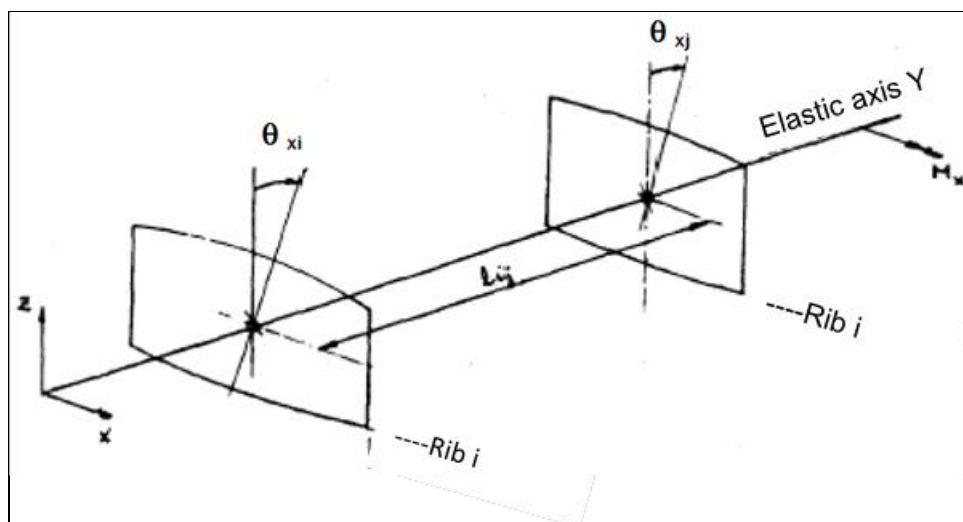


Figure 41. Elastic Axis definition and rotation

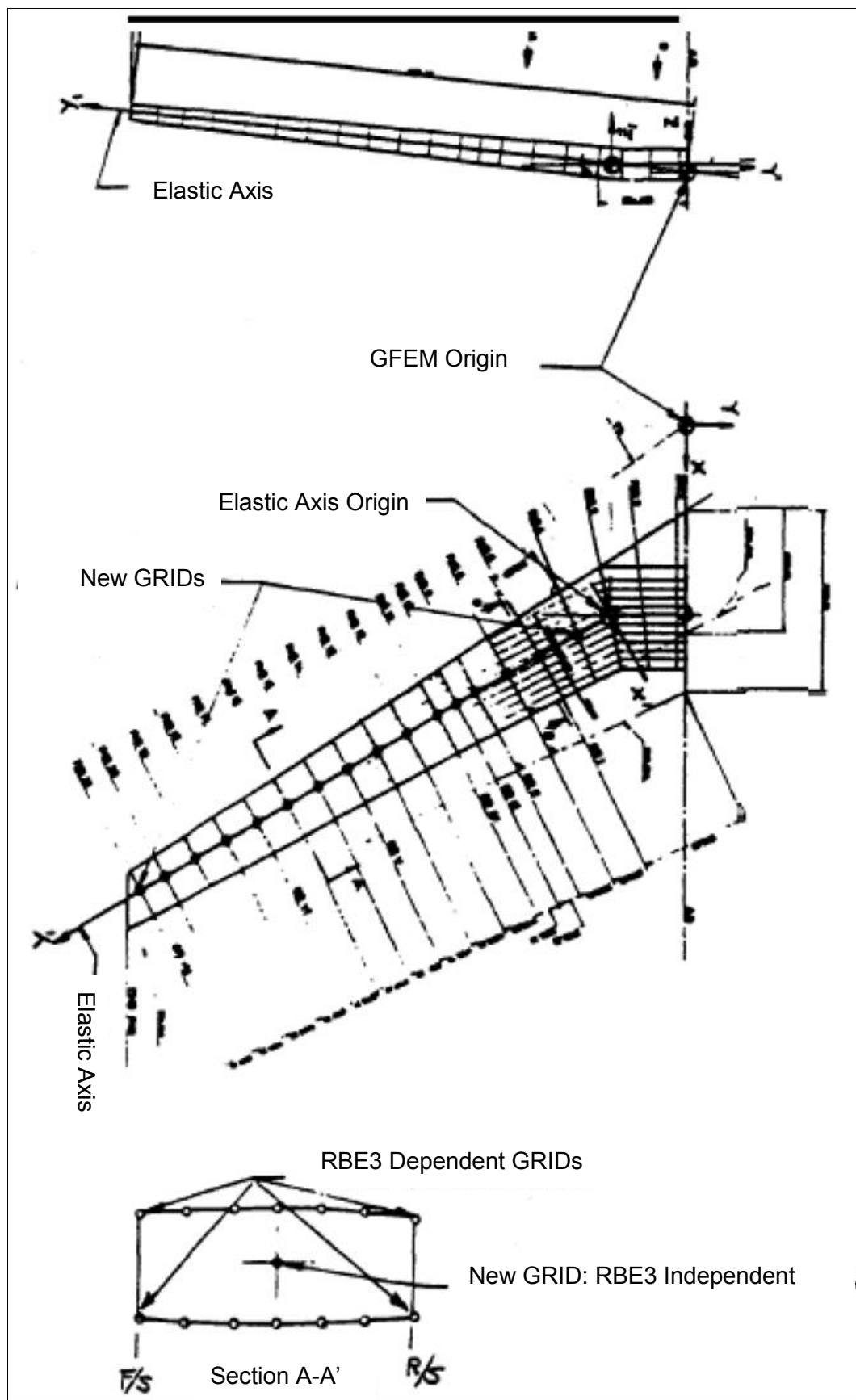


Figure 42. Elastic Axis and RBE3 connection

10. ANNEX

10.1 Composite Shell Modelling

The idealization of composite panels deserves a specific treatment due to the special characteristics that these materials present in terms of constitution and anisotropy.

10.1.1 Solid Laminate Shell

The elements to be used in its idealization will be CQUAD4 with

- Membrane properties for thin shells
- Shell elements for thick shells
- or only with shear properties (G33), so far, the same as metal panels.

The way to model the mechanical properties of the element and the properties of the material show the difference regarding to the metal panel. In the case of a composite panel, the QUAD4 elements will refer to PCOMP and MAT8 cards or, when it is necessary to supply models to external companies, the QUAD4 elements will refer to PSHELL and MAT2 cards so, it will be necessary to calculate the membrane matrix of the element (and in some particular cases the flexure and perhaps the transverse shear), taking into account the angle between the axis of the element and the direction of the fibre.

When the thicknesses are very high, the effects of bending and offsets are taken into account (this idealization is not valid for buckling or non-linear studies, for these cases the position of the nodes must be modified by placing them in the middle line of the elements).

Before continuing, it is convenient to clearly define a series of concepts. The plane of the nodes will be the one that contains them and does not have to coincide with the plane of the element (reference plane) that will be linked to the element used (and which is set when the element is defined using the ZOFF parameter) nor with the average plane of the element. As we have been stated throughout this document, the nodes will generally be placed on the outer surface of the element to be idealized. On the other hand, note that in the two options considered below, the bending stiffness of the idealized element and the layers of the stack are listed starting with the most negative z-coordinate. The two possible options are:

1. The first option (see Figure 43) consist in using the offset parameter (ZOFF) of the QUAD4 elements assigning the value $\pm t/2$ (remember that it is not the default value of this parameter) to define the element plane (plane of reference) to idealize (minus sign if the plane of the element is below the plane of the nodes, taking into account the connectivity). For this option (the Zo parameter of the PCOMP card will be left blank (taking its default value that is $-t/2$) so that the mid plane of the element will match with the reference plane.

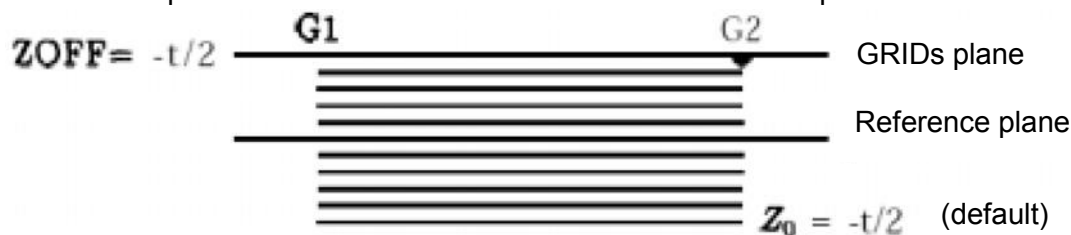


Figure 43. Option 1 for modelling composite shell in MSC.Nastran

2. In the second option (see Figure 44) the parameter ZOFF will take its default value (which is zero and therefore it will be left blank) so that the plane of the nodes and the reference plane will match, and will be the parameter Zo of the PCOMP card who adopts the offset value in the usual case of nodes located on the outer surface of the element (minus sign if the plane of the element is below the plane of the nodes, taking into account the connectivity).

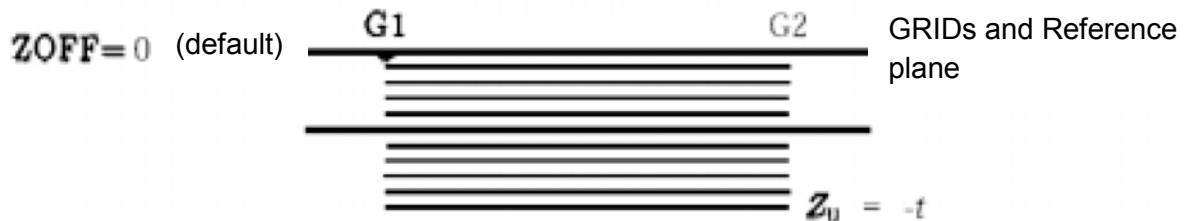


Figure 44. Option 2 for modelling composite shell in MSC.Nastran

The two options are totally equivalent in terms of overall results in the linear solver, but the following peculiarities have to be taken into account:

- The output of forces and strains in the second option is referred to the plane of the nodes (which matches with the reference plane in this option) and not the middle plane of the element. Therefore moments applied in these nodes will appear.
- The stress output in $\pm t/2$ when the Z0 parameter of the PCOMP card is different from the default (second option), are not correct. It is necessary to use the first option if you want to directly assess these values.
- The rest of the outputs are coincident in both options, including the outputs layer by layer.

Regarding the previous points, the advantage of the first option over the second one seems clear, but it has the disadvantage of linking the thickness of the laminate given in the PCOMP card with the definition of the QUAD4 card (value of the ZOFF parameter), which can induce many times to error in model modifications.

In any case, the PCOMP and MAT8 cards are internally transformed by the MSC.Nastran in equivalent PSHELL and MAT2 cards, operating with the latter. MSC.Nastran gives in the output file the equivalent cards to the PCOMP and MAT8 cards supplied. Note that these PSHELL cards include the bending stiffness of the idealized element. In the case that, for any reason (very thin panel or due to structural element hypothesis) it is not adequate to attribute this property to the element, it can be written in the field 9 of the PCOMP card ("LAM") the value "MEM" (see Table 12), the PCOMP and MAT8 cards can also be replaced in the input file by their PSHELL and MAT2 equivalents (obtained from a previous execution) where the bending property has been eliminated in the PSHELL cards (blank field for the material in flexion) getting the same effect.

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
CQUAD4	IDQUAD	IDN	G1	G2	G3	G4		-t/2	

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
PCOMP	IDN							MEM	OABC
+ABC	IDMAT	t ₁	θ ₁			t ₂	θ ₂		OEFG
+EFG		t ₃	θ ₃			t ₄	θ ₄		OHII
+HIJ		t ₅	θ ₅						

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
MAT8	IDMAT	E ₁	E ₂	μ ₁₂	G ₁₂				

Table 12. MSC.Nastran cards for modelling composites material only with membrane properties

(Notice that all the layers have to be written although the stack is symmetrical because “MEM” and “SYM” options cannot be selected at the same time)

The parameter PARAM, NOCOMPS, offers the stresses by layer (with the value -1 is obtained the stress in the equivalent homogeneous element, with 1 values are obtained by layer and with the value 0 both are obtained).

10.1.2 Sandwich Panels Modelling

The idealization of sandwich structures will be treated as a particular case of the idealization of composite laminates. Ref. 17 analyses and compares the different options that can be followed to idealize a sandwich element. Basically there are two possible options that differ in the way of entering the properties of the QUAD4 element that represents the sandwich:

- Use PCOMP and MAT8 cards (Table 13). The CQUAD4 card will refer to a PCOMP card with the stack that represent the sandwich, considering the core as one more layer that calls a MAT8 card different from that referenced by the layers that make up the faces of the sandwich. This MAT8 card referenced by the layer defining the core has modules in the plane almost null (10^{-6} , for example) and transverse shear stiffness G1z and G2z. This option considers the sandwich element as a particular case of lamination of composite material with the only difference that the layer defining the core refers to a different MAT8 card to be able to define the properties of transverse stiffness that it possesses. Therefore, within this option, it is possible to use in principle any of the two variants defined in the idealization of laminates of composite material (See 10.1.1: use the parameter ZOFF of QUAD4 or Z0 of the PCOMP card), recommending as there the first of them. Internally, the MSC.Nastran transforms these property cards into equivalent PSHELL and MAT2 cards.
- Use PSHELL and MAT2 cards (see Table 14): In this option, the PSHELL and MAT2 cards that represent the element are built directly, taking advantage of the possibility of the PSHELL card to define a different material for deformation in the plane (membrane), bending and transverse deformation. Thus, two MAT2 cards will be used:
 - A MAT2 card to define the properties of membrane and bending which will be defined by the components A_{ij} of the stiffness matrix in the plane of the laminate formed by the addition of the thicknesses of the faces (it is worth remembering that in MSC Nastran the matrix [A] links stress with strain and not element forces and flows with strains).
 - Another MAT2 card will define the properties of transverse stiffness by writing the transverse shear stiffness G11 and G22 in the fields corresponding to A11 and A22 respectively and leaving the rest of the fields of the card blank.

The definition of the PSHELL card is completed by stating that:

- The membrane thickness is the addition of the thicknesses of the sandwich faces, $2t$.
- The bending stiffness parameter ($12I/T3$) has the following value where “c” is the thickness of the core and “t” is the thickness of one of the faces:

$$\frac{1}{4} \left[1 + 12 \cdot \left(\frac{c+t}{2t} \right)^2 \right]$$

- The ratio of transversal shear thickness and membrane thickness is $c/2t$.

When the mid plane of the element does not match with the plane of the nodes, the ZOFF parameter of the CQUAD4 card will be used to define the position of this middle plane of the element (take into account that the positive distance is determined regarding to the plane of the nodes following the rule of the right hand and the order in which the nodes are given in the card CQUAD4). This situation can also be solved by defining a new material that represents the bending-membrane coupling in the plane of the nodes but involves laborious calculations that usually lead to a bad conditioning of the element's stiffness matrix. These calculations are greatly complicated when the sandwich to be idealized is not symmetrical. Therefore, this possibility will not be used.

The simplicity of the first option regarding to the second is evident but presents a slight drawback: The stress output request on the upper and lower faces is not correct because they are calculated with the hypothesis that the entire section is resistant to axial stresses, hypothesis that is not true because of the presence of the core.

In the other hand, the stress output by layer is correct in this first option. Thus, if it is desired to assess the stresses on the faces ($\pm t/2$), it will be necessary the second option or calculate these values externally from others.

In any case, the first option must be used with priority to the second, as indicated for the idealization of composite laminates. It should be remembered, as we did for solid laminates, that for buckling as well as non-linear studies, the plane of the nodes should be in the midline of the element avoiding the use of offsets.

The parameter “PARAM, NOCOMPS, 1” offers the stresses layer by layer (The value -1 gives the stresses in $\pm t/2$ and with the value 0 both are obtained).

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
CQUAD4	IDQUAD	IDN	G1	G2	G3	G4		-t/2	

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
PCOMP	IDN		θ_1					SYM	OABC
+ABC	IDMAT	t_1	θ_3		IDMAT	t_2	θ_2		OEFG
+EFG	IDCORE	$t_{CORE}/2$							

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
MAT8	IDMAT	E_x	E_y	μ	G				

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
MAT8	IDCORE	1E-6	1E-6			G_{12}	G_{22}		

Table 13. Sandwich idealization with PCOMP, MAT8 and CQUAD4 cards

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
CQUAD4	IDQUAD	IDN	G1	G2	G3	G4		-t/2	

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
PSHELL	IDN	IDMAT	t ₁ + t ₂	IDMAT	I	IDSH	c/2t		OABC
+ABC	-(c/2+t ₁)	-(c/2+t ₂)							

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
MAT2	IDMAT	A ₁₁	A ₂₂	A ₁₃	A ₂₂	A ₂₃	A ₃₃		

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
MAT2	IDSH	G ₁₁	0	0	G ₂₁				

$$I = \frac{1}{4} \left[1 + 12 \left(\frac{c+t}{2t} \right)^2 \right]$$

Table 14. Sandwich idealization with PSHELL, MAT2 and CQUAD4 with offset

10.2 MPC's and Rigid Body Elements

Missions of MPC's and RBEs:

- Provide a correct structural stiffness to nodes in any or all degrees of freedom in which they present:
 - Null stiffness in some degree of freedom and avoiding the use of SPC when preventing the displacement.
 - Very low stiffness that produces numerical singularities in the stiffness matrix (MAXRATIO greater than 10⁷), and that if its movement is prevented with an SPC, appreciable reaction forces/moments are produced.
- Connect nodes whose stiffness/displacements are not important (dummy nodes) but in which there are applied loads that must be transmitted to the structure.
- Connect "generalized" nodes to a structure. A "generalized" node is one in which masses corresponding to an area of the structure are applied, and this node is intended to move as the average of a section. These nodes are normally used to:
 - Apply the inertial loads associated with that mass.
 - Perform verifications of rigid solid (see Ref. 8), supporting the structure in that node (usually the centre of the most representative section of the structure is chosen) and imposing forced displacements in the six degrees of freedom of that node.
 - To be representative of the movement of a section (and to have assigned the mass associated to that section) in dynamic analysis.

As a general rule we should try to avoid the use of rigid elements due to the complexity and its difficult post-processing, although there are situations that make its use essential.

It should be noted that the post process of the rigid elements is performed through the output of the balance of forces in the nodes (GPFORCE): the line that represents F-OF-MPC (Nastran output) is the force that the elements whose are connected apply towards the node. When several rigid elements use the same node, it is impossible to assess what part of the force corresponds to each rigid element.

Before listing some of the most typical uses of rigid elements in a FEM, it should be clear that the entire rigid element is equivalent to a MPC ratio, that is, an imposition in the mathematical model of additional equations that match dependent degrees rigid of freedom (set M) with other independent

(set N). It is important to note that as those dependent degrees of freedom are deleted from the equations, they cannot be part of the set of analysis (set A), nor be restricted its movement by SPC, nor of course be part of another MPC/Rigid element as dependent. Neither can dynamic variable loads be used, nor can forces or stresses be recovered in them.

Some differences between an MPC and a Rigid Element:

- in the MPC, the user chooses the coefficient that applies to each DOF and in the Rigid Element is MSC.Nastran which calculates them by combining averaging coefficients deduced from the geometry.
- MPC can be called or not according to Case Control commands. RBEs are always active.

It must always be taken into account that the set of independent degrees of freedom that is chosen in the rigid element must be enough for preventing displacements as rigid body.

In the same way, it is preferable not to establish excessive or unwanted constraints that imply restrictions in the deformation shape.

As the general advice it is recommended that all degrees of freedom linked and in a rigid element or MPC are defined with the Analysis same coordinate system (Field 7, GRID Card) to avoid errors, although there are situations in which that is used as a trick (see in the explanation in section 7.2. a way to idealize moving surfaces at different angles).

It is needed a perfect comprehension of the correspondence between the Rigid Element and its mathematical equations and to forget the wrong idea that a Rigid Element is always an element of infinite stiffness in which some nodes are tied, especially in the MPC

Below are the most frequent uses of MPC and rigid elements.

10.2.1 Explicit MPC

The MPC card use is mainly to impose the movement of a node in a certain degree of freedom, associating it, through coefficients, with others. Respond, therefore, to the equation:

$$A\delta_1 + B\delta_2 + C\delta_3 + \dots + (N-1)\delta_{n-1} + N\delta_n = 0$$

Or what is the same and more generally:

$$U_M = - \sum_i f_i U_{Ni}$$

U_M is de dependent DOF and the U_{Ni} are the independent DOFs.

Two typical usages of MPC's have been explained in chapter 7. Another usage of MPC's is in the Nastran 'Linear Gap' simulation (see Ref. 6).

The MPC's can be active by Nastran Case Control MPC cards and MPCADD in Bulk data section.

10.2.2 RBAR

The RBAR element is a Rigid Body Connection that has three fundamental utilities:

1. A Rigid-BAR connection between 2 GRID nodes where the translational and rotational degrees of freedom can be linked.
2. To connect "generalized" nodes with each other: for example in a "motor dummy" model, the aerodynamic nodes of the Cowling are connected by RBE2 to the CG of the Cowling (which remains in set N in this construction) and the latter by means of RBAR to the CG engine (remains in the set M the first), then joining all this "structure" dummy "to the actual structure of the aircraft, by connecting that engine CG to the" structural "model (for example with RBE1 or RBE3).
3. In the use of the Technique of External Super-elements, when duplicating the interfaces between components, those duplicate nodes, GA and GB, must be joined by RBAR in the six degrees of freedom, introducing as dependents the degrees of freedom from the internal node to the superelement, GB

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBAR	EID	GA	GB	123456			123456		

Table 15. MSC.Nastran RBAR card

10.2.3 RBE1

This is a Rigid Body element used for its simplicity to connect degrees of freedom without stiffness. It allows connecting an arbitrary number of points: the dependent degrees of freedom (set M) and independent (set N) can be chosen arbitrarily by the user.

Some application example of RBE1 usage is to connect different LIM nodes (non-structural nodes) to structural FEM.

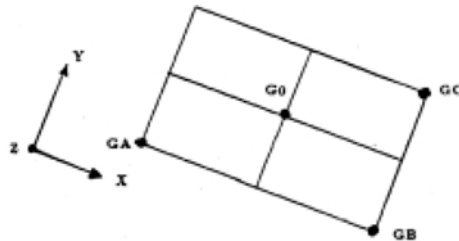
As independent (set N) the number of degrees of freedom to include must be six and only six (capable to define a Rigid body). A simple mnemonic rule to select the independent DOSs:

"3 translational degrees of freedom are chosen from a node (123), which is capable of supporting them (which has stiffness in those degrees of freedom). Then one moves to the second chosen node, eliminating the degree of freedom in which we have moved (the most predominant if we have not moved according to the axis): for example, if we have moved along the Y axis, we remove the 12, leaving only the 13. Then we move from the second node to the third, eliminating the predominant degree of freedom in which we move: that is, if we now move in a predominantly Z direction, we would eliminate the 3, leaving only 1 for the third node".

These six independent DOFs define a Rigid body in which we can 'hang' or 'link' whatever GRID or DOF with no stiffness (as a LIM node), see Table 16 or Table 17.

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE1	EID	GA	123	GB	23	GC	3		OABC
+ABC	UM	123456	GO						

Table 16. RBE1 Nastran card. 6 DOFs dependents



- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE1	EID	GA	123	GB	23	GC	3		OABC
+ABC	UM	3	GO						

Table 17. RBE1 Nastran card for constraining normal DOF in shells

10.2.4 RBE2

RBE2 is a real Rigid Body Elements. The definition identifies some GRID nodes that will move according to displacements and rotations of the independent GRID. Two typical applications:

- Represent a big block or a high stiff part or equipment.
- The connection of an arbitrary number of non-structural (dummy) nodes in which loads are applied, to a single structural node in its six components. It has the limitation that all non-structural nodes are connected in the same set of degrees of freedom (not some in some degrees and others in others).

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE2	EID	GO	123456	GA	GB	GC	GD	GE	OABC
+ABC	GF								

Table 18. MSC.Nastran RBE2 card

10.2.5 RBE3

It is the most common element due to its utility, but it can reach a greater degree of complication. In general, any of the utilities of a RBE1 can be performed with a RBE3, but with greater complication of the card. Another big difference is that, in general, it does not affect the stiffness matrix. It can be used to avoid lateral singularities in fittings with acceptable static and dynamic results.

RBE3 is the Nastran element used to distribute loads and masses to the GFEM structural nodes.

The RBE3 defines an attachment relationship in which the movement of a "reference point" is determined as the movement of another series of nodes, affected by weighting coefficients that the user directly enters in the card.

The most typical use is the distribution of nodes/masses ("beaming" in MSC.Nastran terminology) applied at the "reference point" to a set of nodes, that is, to connect a defined point in the centre/reference point of a section in which:

- A concentrated mass is defined (CONM2)

- A load associated with that section will be applied (for example, the inertial load corresponding to the mass of the previous section), distributing it between the contour nodes according to weighting coefficients.
- The model will be fixed for a rigid solid check by applying forced displacements in the six possible directions to that node, and checking how the rest of the model moves.

The recommended construction of the card will be as follows, leaving the six degrees of freedom of the reference node as dependents (in set M) and using translational DOFs (1, 2 and 3) as independent for the nodes used for connection:

- 1 -	- 2 -	- 3 -	- 4 -	- 5 -	- 6 -	- 7 -	- 8 -	- 9 -	- 10 -
RBE3	EID	-	GO	123456	L	123	GA	GB	OABC
+ABC	GC	GD	GE	GF					

Table 19. MSC.Nastran RBE3 card

Some rules has to be followed to obtain good results from RBE3:

- Try to avoid (if possible) the use of rotational DOFs for the independent GRIDs
- The reference GRID (G0) should be almost centered related the 'weighted' center of the independent GRIDs, to avoid non-logical load distributions.

10.3 Use of External Superelements Technique

The Superelements technique is a utility of the MSC.Nastran that allows handling large models of finite elements, or those in which several companies work and each one wants to keep its part of the structure "hidden" from the others to keep data safe that could be confidential. In "huge" models we can have problems with the limitation of memory of the computer system (hardware). This can be avoided with this technique.

For more complete information about MSC.Nastran Superelement technique, please take a look at the Ref. 12.

The use of super elements involves the partition of a large structure into a series of substructures (called superelements) that can be processed separately or together.

Disadvantages:

- Requires more theoretical and practical knowledge of the MSC.Nastran program.
- Sometimes it takes more work to complete the complete model partition and form and define the Superelements.
- Need for more disk space for permanent databases.
- Increase in the CPU time required for the initial execution.
- The MSC.Nastran documentation on this subject is sometimes incomplete, generally in relation to external super elements.

By cons, the advantages are:

- Each superelement is a free structure in itself: it can be verified and validated more easily in a rigid solid check or modal analysis. Each Superelement is processed separately.
- Less number of operations (saving CPU time).
- The required RAM space is smaller.
- Only the affected super elements and the residual structure need to be aligned before a change in the model.

For very large models, problems with the limitation of memory of the computer system or joint execution of a model among several partners may appear so, usually, the advantages largely counteract the disadvantages. Otherwise, the disadvantages usually weigh more than the advantages of their use. And in any case, of the two possible ways of internal management of the superelements, external and internal, external superelements will always be used for the clearer physical information they provide.

The Superelement technique can also have other applications: it can be used in structures with some type of symmetry by means of the image method (see Ref. 9). The final goal of this method is to reduce the idealization time and the size of the problem (KAA stiffness matrix), generating nodes, connectivity, properties, materials and MPCs for only one part of the structure (which will be assigned to an internal superelement), reproducing then that part in another place, identically or applying symmetry according to a plane or an axis, or even making a mirror image or identical of the model according to the three axes.

From the recent versions there are two ways to define the superelements in NASTRAN: the traditional method, called "Main Bulk Data Section", which defines the superelements within a single bulk data through "SESET" and others, and the new method, denominated Superelements by Parts, in which they redefine the superelements in bulk data separately (BEGIN SUPER command), that allows a very simple definition and the connection is made of automatically. For more details, see Ref. 9

A very useful application is the generation and management of external super elements, the reduction of models to a matrix and their use in a general model.

It consists of two steps:

- Step 1: Reduction of the external superelement
- Step 2: Addition of external superelement and analysis

10.3.1 Procedure for reducing NASTRAN models

Input form: It is a normal execution in which the following cards are used:

- EXTSEOUT: card used in the case control for the overturning of the superelement
- ASET, ASET1 to select contour nodes that will be used in the reduction.
- Other cards related to nodes (BNDFIX, BNDFIX1, BNDFREE, BNDFREE1, BSET, BSET1, CSET, CSET1, QSET, QSET1)

EXTSEOUT: This card has the following format:

EXTSEOUT [(STIFFNESS, MASS, DAMPING, K4DAMP, LOADS, {ASMBULK}, {EXTBULK}, {EXTID = seid}, {MATRIXDB, DMIGDB, DMIGOP2=unit, DMIGPCH})]

The parameters in bold are dumped by default:

- I. STIFFNESS: if activated, output the reduced stiffness matrix
- II. MASS: if activated, output the reduced mass matrix
- III. DAMPING: if activated, output the reduced viscous damping matrix
- IV. K4DAMP: if activated, output the reduced structural damping matrix
- V. LOADS: if activated, output the static charge matrix reduced
- VI. ASMBULK: create the assembly cards in a *.asm file. The data are: SEBULK, SECONCT, Contour nodes and CORD2x associated with the previous ones.
- VII. EXTBULK: generates the cards used in the corresponding portion of the associated BEGIN SUPER. This card is ignored if DMIGPCH is specified. Contour nodes are generated, interior nodes referenced by PLOTELS, CORD2x associated with all the above, EXTRN, ASET/ASET1, QSET/QSET1, SPOINT and PLOTEL
- VIII. EXTID=seid: specifies the superelement index to be used in ASMBULK or BEGIN SUPER. It must be specified if ASMBULK, EXTBULK or DMIGPCH is used.

Superelement Format options

- MATRIXDB: write contour matrices and other information.
- DMIGDB: same as the previous one but with the difference of using DMIG format.
- DMIGOP2=unit: same as the previous one but write the results in an OUTPUT2.
- DMIGPCH: writes the arrays in a *.pch file.

Reduction execution

To configure a reduction execution, it is recommended to use the EXTSEOUT function with the minimum necessary parameters, since it can be used on an existing model with a minimum of changes. This means that if only stiffness is needed, the keyword would be invoked. The DMIGPCH command is also recommended as a dump option. This command is available since MSC.Nastran version 2004, so it cannot be used with previous versions.

Example of the command:

EXTSEOUT (STIFFNESS, LOAD, EXTID=10, DMIGPCH)

Depending on the type of reduction to be used, one solution or another should be used.

- Static reduction: SOL 101, with the load cases necessary to reduce.
- Dynamic reduction: SOL 103, with the load cases necessary to reduce. It will be necessary to use loadset and lseq cards to apply the necessary loads. It is recommended to use the residual vectors method to improve the reduction (param,resvec,yes).

If the number of load cases is high and can be composed based on unitary load cases, it is advisable to reduce only these and combine the rest using LOAD cards. If the number of cases is high, but the number of load application points is small, those of load application can be included as contour nodes, in case the loads do not need to be reduced. This last case adds another advantage, which is the possibility of including load cases after the reduction without the need to re-execute or modify any card.

In the Bulk data it will only be necessary to add the ASET/ASET1 cards necessary to define the degrees of freedom of the contour nodes that we want to keep.

If it is desired to be able to vary the configuration of SPCs or MPCs of the model, all the nodes affected by these conditions must be retained, in order to maintain the corresponding cards and be able to use them from the CASE CONTROL.

In case of containing the model some non-linearity (CGAPs for example), these should be outside the reduction and added to it later.

Reporting the results of the reduction. The command, when executed, shows the following information:

- Data related to the matrices: format, writing medium and identification.
- Bulk data output: contour or contour and plotel.
- Type of reduction: static or modal condensation.
- List of fixed contour nodes.
- List of free contour nodes.
- List of generalized coordinate nodes.

Outputs

ASMBULK	EXTBULK	DMIGPCH	OUTPUT
			Without output
X			*.asm to be used in MAIN BULK DATA
	X		Data written in the standard punch file for being used in the corresponding BEGIN SUPER
		X	
X	X		Two separated parts. One to be used in the MAIN BULK (*.asm) and the other for the corresponding BEGIN SUPER
X		X	

Table 20. Output files from a SE run

10.3.2 Addition of the Superelement and execution

Two parts are differentiated into a model divided into super elements:

Main Bulk Data: will contain only the part of the model belonging to the residual and data related to the assembly of the superelements.

Super Element Bulk Data: will contain only the data of the superelements dumped by EXTBULK or DMIGPCH (recommended method).

In the case of not want to add the superelement as external, there is always the possibility of including it as part of another model.

This would be done by removing all parts referring to superelement from the output file (such as ASET/ASET1 and BEGIN SUPER) and using the commands K2GG, M2GG, B2GG, K42GG and P2GG. The name of the matrices dumped by default are KAAX, MAAX, BAAX, K4AAX and PAX; which it is recommended to replace with another name that better matches the description of the model.